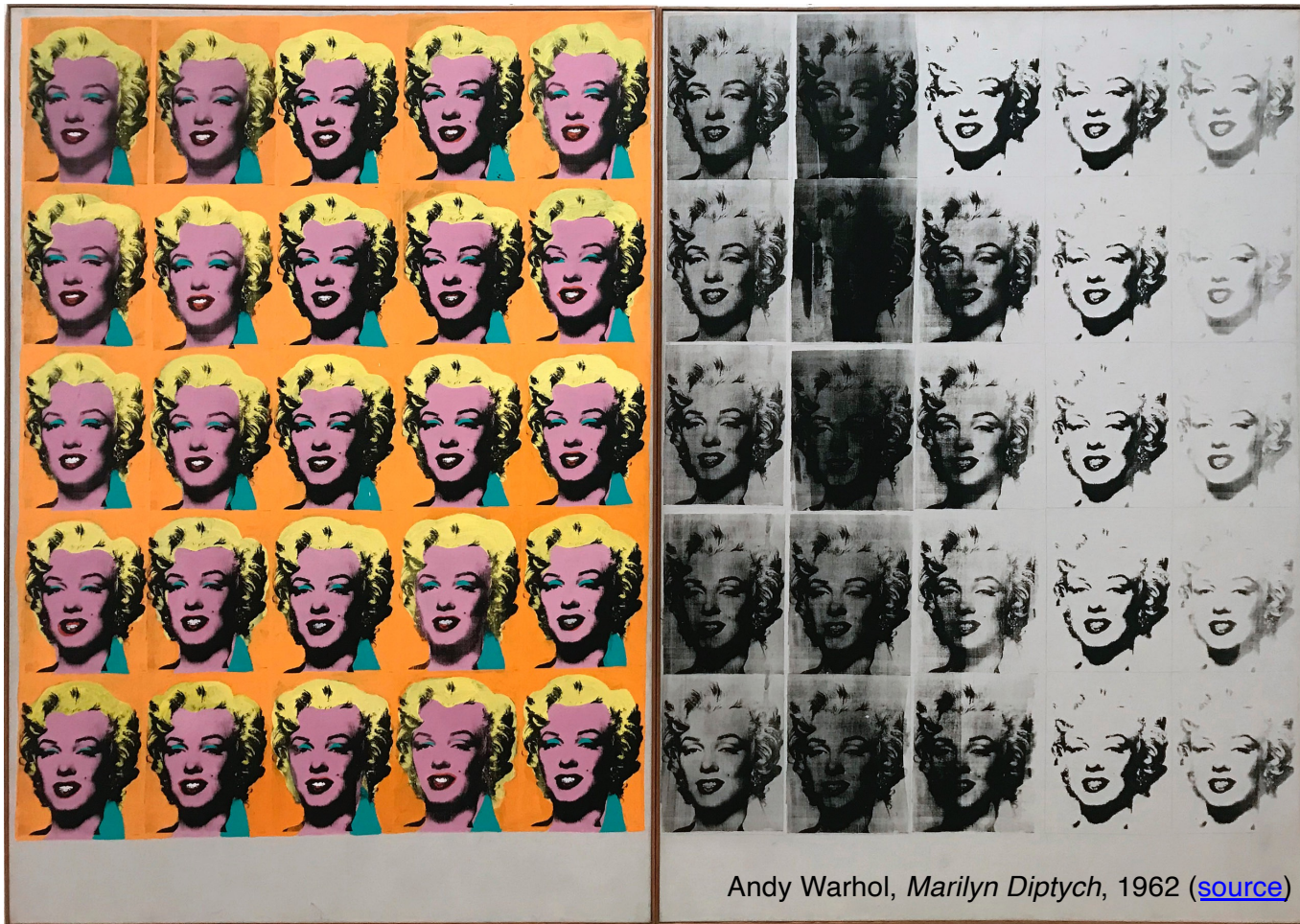


Image processing basics



Andy Warhol, *Marilyn Diptych*, 1962 ([source](#))

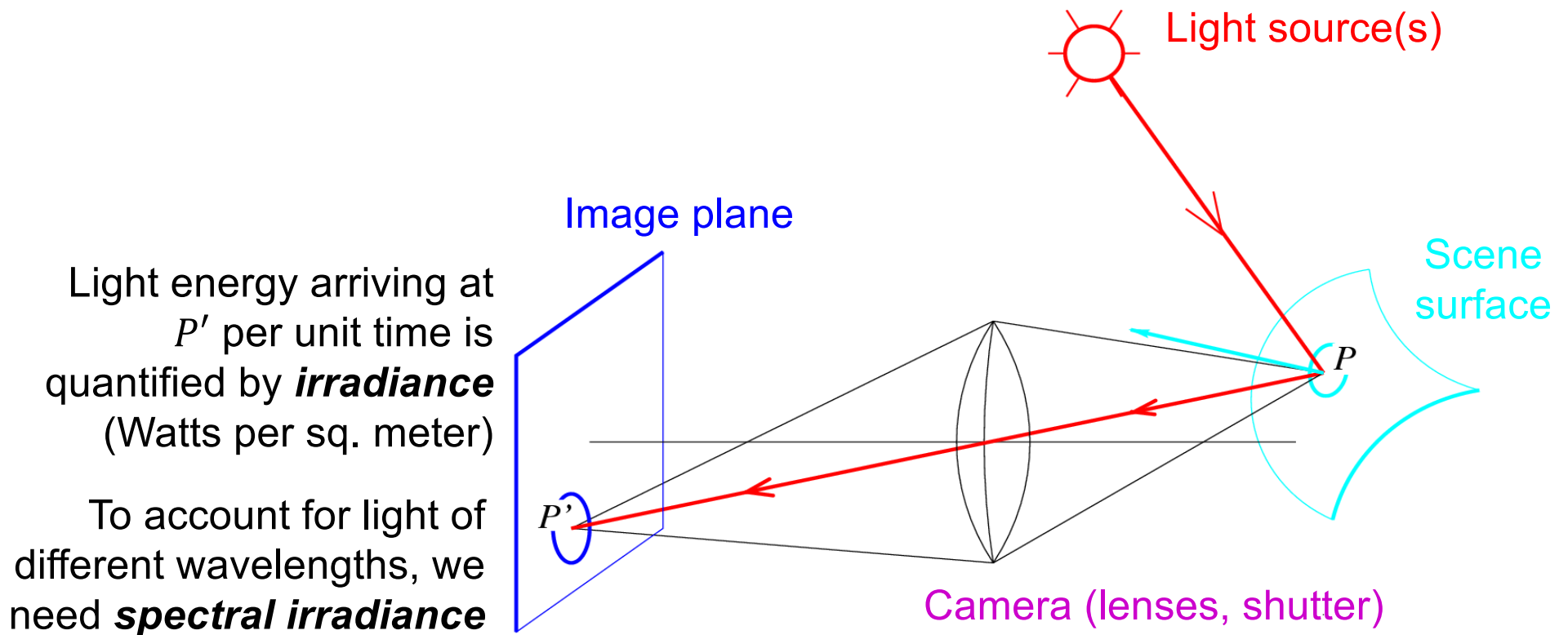
Many slides adapted from
[Alyosha Efros](#), [Derek Hoiem](#)

Image processing basics: Outline

- Images as sampled functions
- Sampling and reconstruction, aliasing
- Image resampling, interpolation
- Image transformations

Image formation (preview)

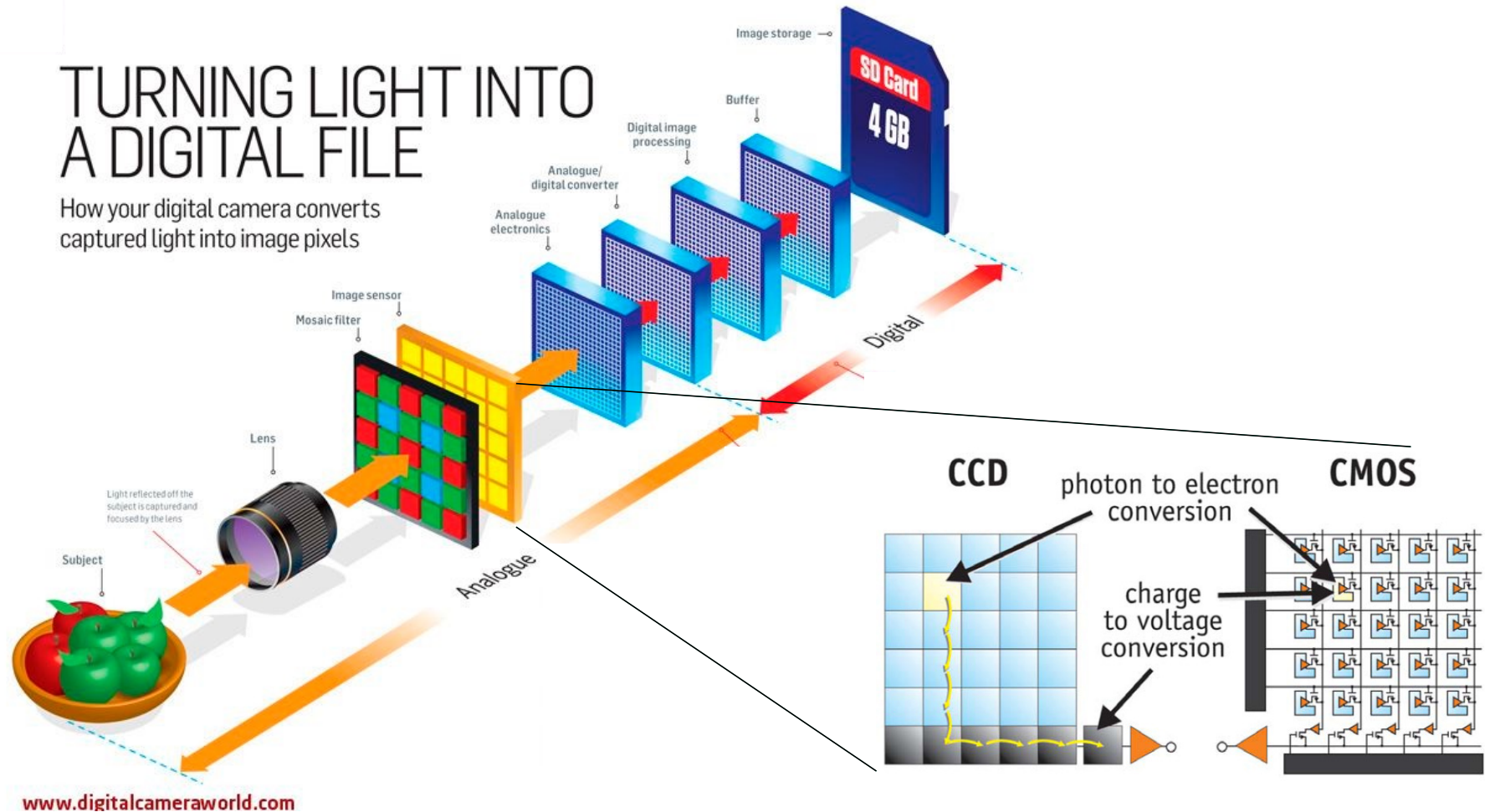
- What determines the brightness of an image pixel?



Images as sampled functions

TURNING LIGHT INTO A DIGITAL FILE

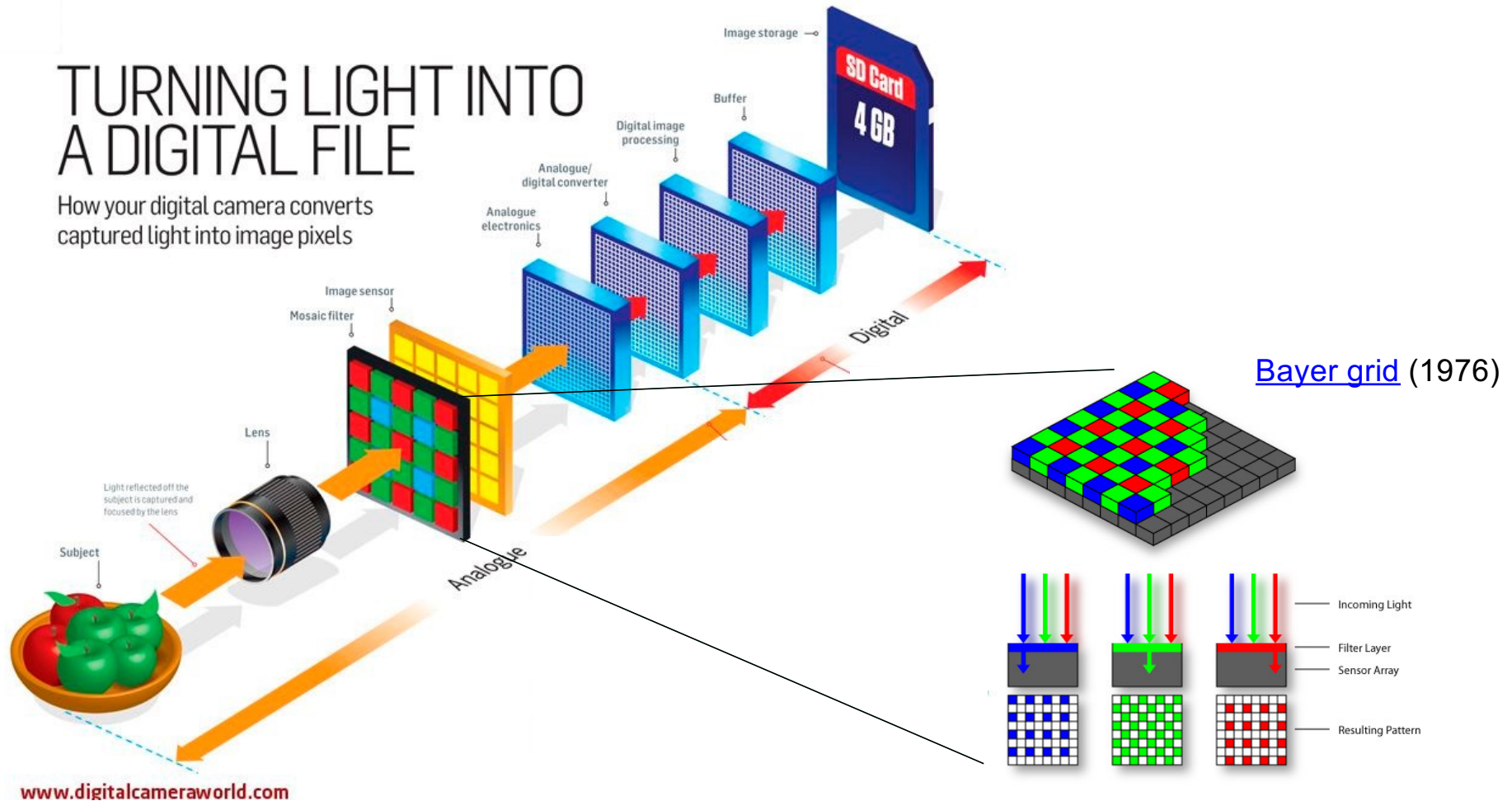
How your digital camera converts captured light into image pixels



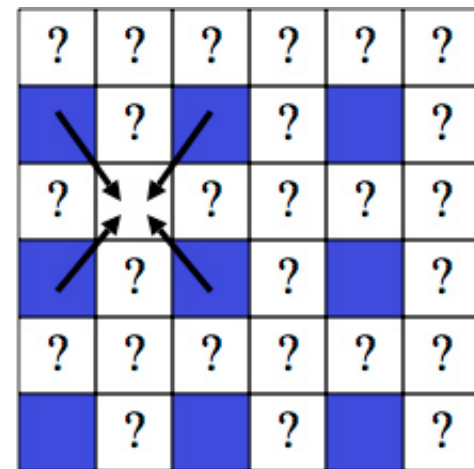
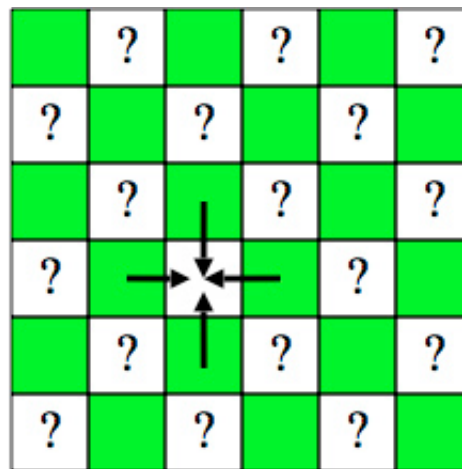
Images as sampled functions

TURNING LIGHT INTO A DIGITAL FILE

How your digital camera converts captured light into image pixels



Demosaicing: How to fill in the missing values?



Digital color images



Digital color images in Python

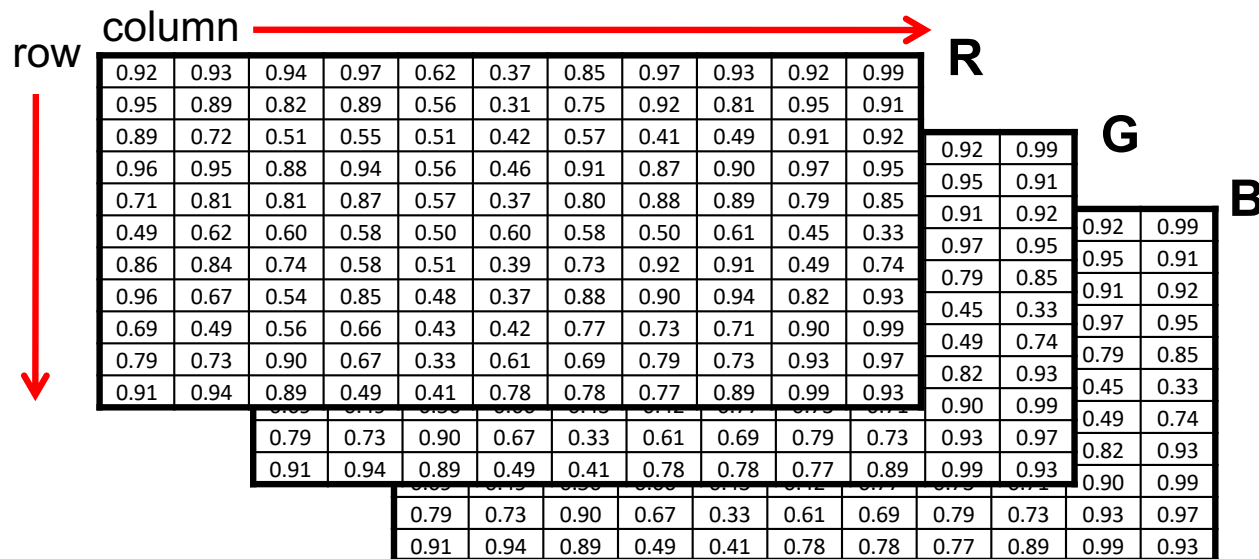
```
im = cv2.imread(filename) # read image
im = cv2.cvtColor(im, cv2.COLOR_BGR2RGB) # order channels as RGB
im = im / 255 # values range from 0 to 1
```

RGB image `im` is a `H x W x 3` matrix (numpy.ndarray)

`im[0,0,0]` is the top-left pixel value in R-channel

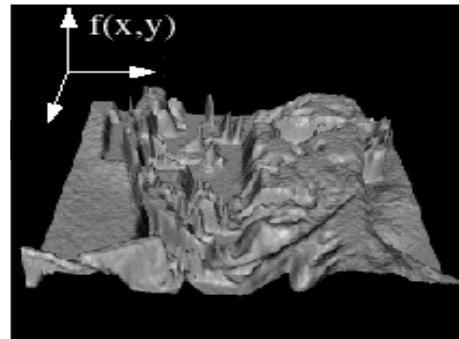
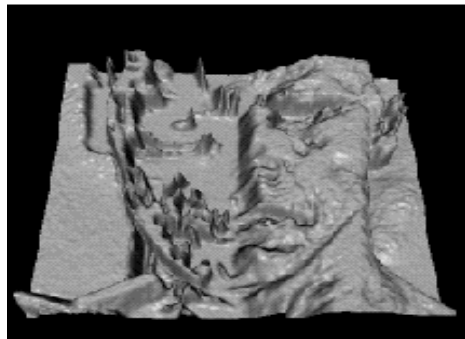
`im[y, x, c]` is the value `y+1` pixels down, `x+1` pixels to right in the `cth` channel

`im[H-1, W-1, 2]` is the bottom-right pixel in B-channel



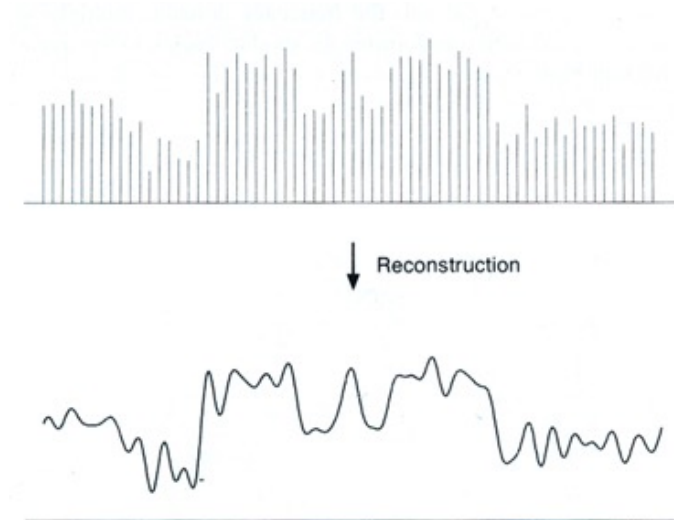
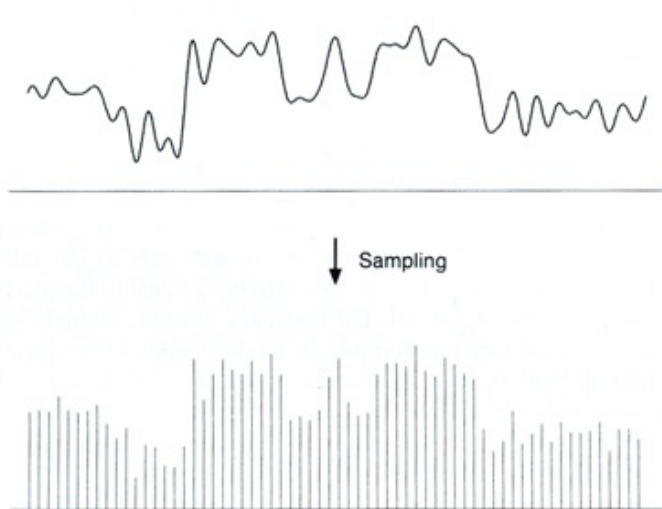
Images as sampled functions

- We like to think of a digital image as a sampled representation of a continuous function $f(x, y)$ defined over a continuous 2D domain



Sampling and reconstruction

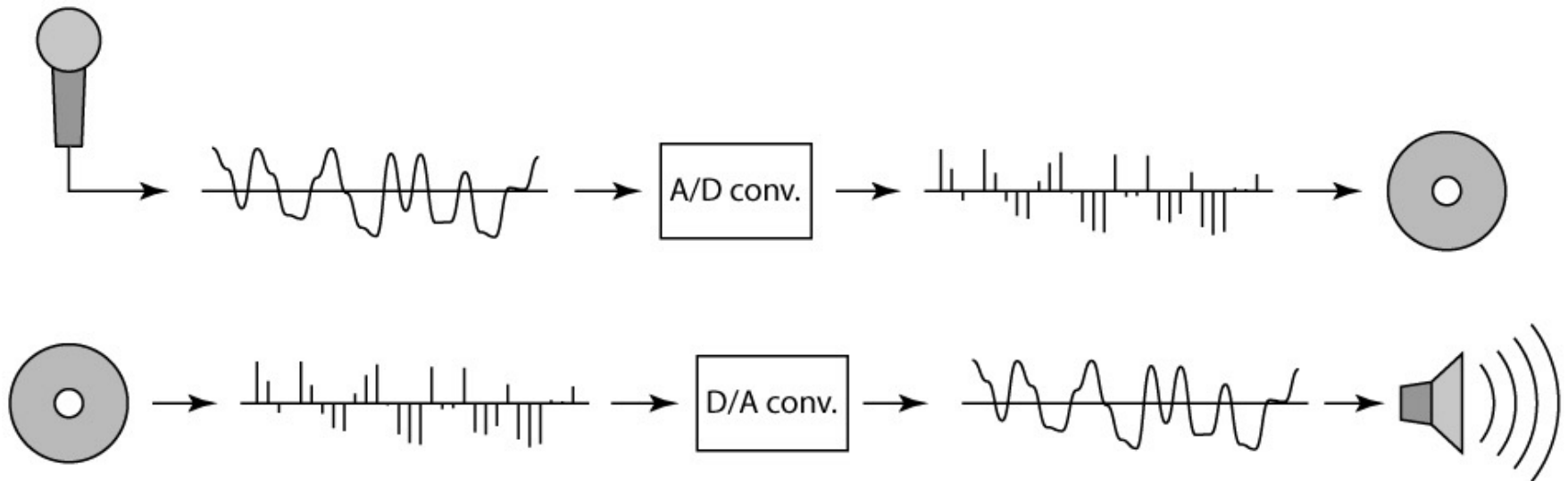
- Sampling: recording the function's values at a discrete set of locations
- Reconstruction: converting a sampled representation back into a continuous function by “guessing” what happens between the samples



Source: S. Marschner (via A. Efros)

1D example: Digital audio

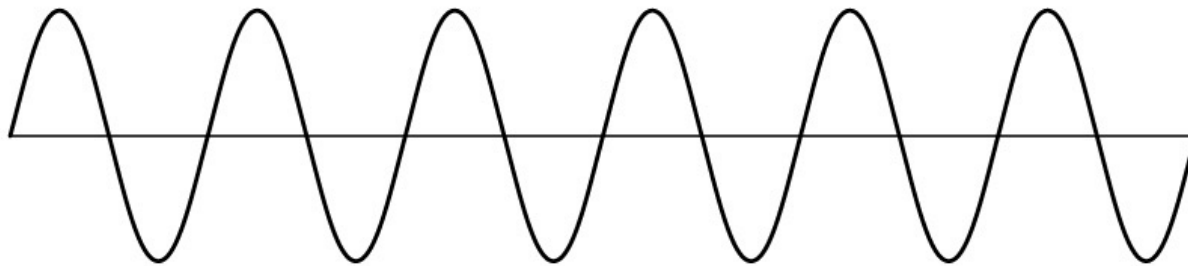
- Recording: sound to analog to samples to disc
- Playback: disc to samples to analog to sound again



Source: S. Marschner (via A. Efros)

Sampling and reconstruction

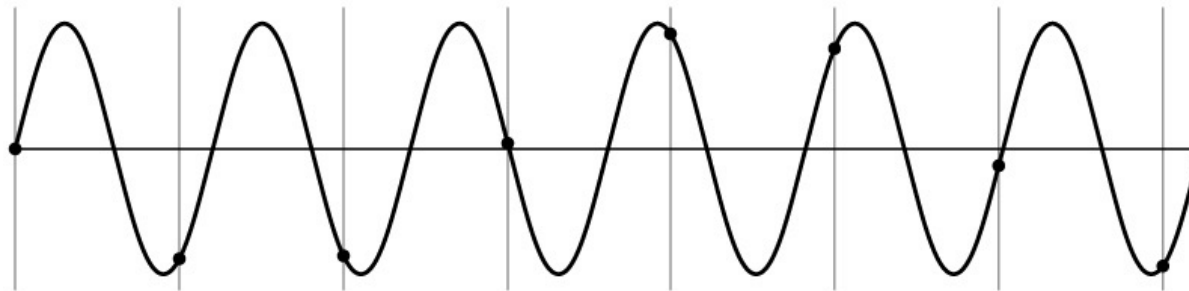
- Simple example: a sine wave



Source: S. Marschner (via A. Efros)

Sampling and reconstruction

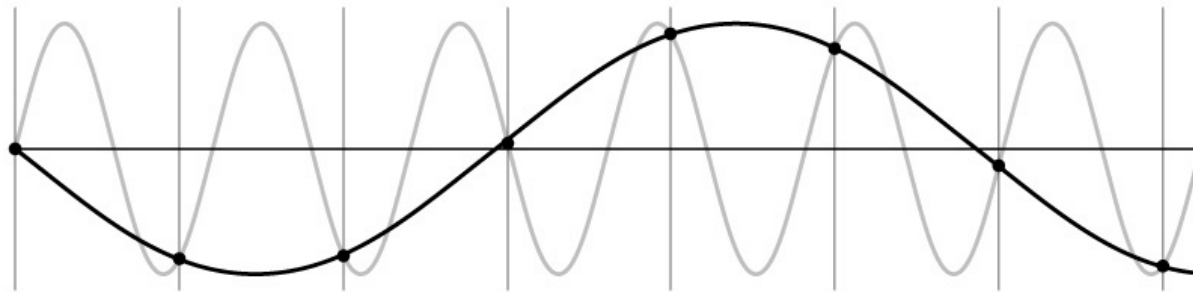
- Simple example: a sine wave
- What if we “missed” things between the samples?
 - Unsurprising result: information is lost



Source: S. Marschner (via A. Efros)

Sampling and reconstruction

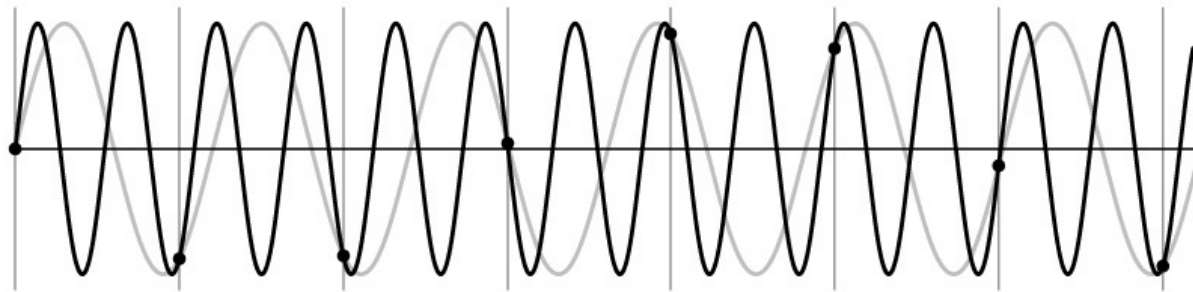
- Simple example: a sine wave
- What if we “missed” things between the samples?
 - Unsurprising result: information is lost
 - Surprising result: indistinguishable from lower frequencies



Source: S. Marschner (via A. Efros)

Sampling and reconstruction

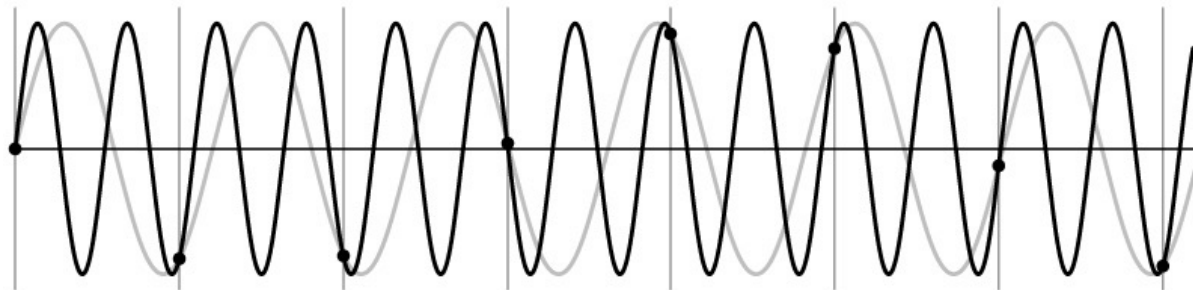
- Simple example: a sine wave
- What if we “missed” things between the samples?
 - Unsurprising result: information is lost
 - Surprising result: indistinguishable from lower frequencies (or even higher frequencies)



Source: S. Marschner (via A. Efros)

Sampling and reconstruction

- Simple example: a sine wave
- What if we “missed” things between the samples?
 - Unsurprising result: information is lost
 - Surprising result: indistinguishable from lower frequencies (or even higher frequencies)
 - *Aliasing*: signal “traveling in disguise” as other frequencies



Source: S. Marschner (via A. Efros)

Wagon wheel effect



<https://www.youtube.com/watch?v=6XwgbHjRo30>

Aliasing in images

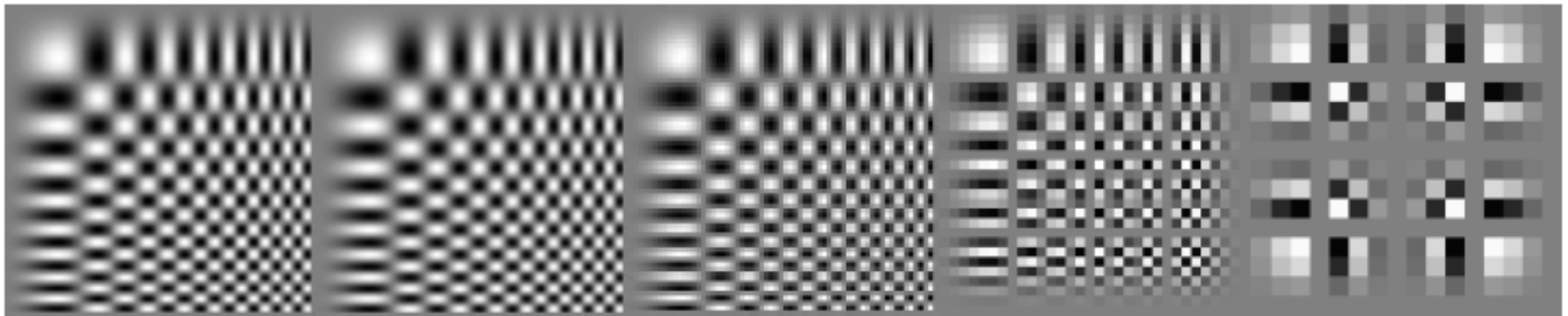
256x256

128x128

64x64

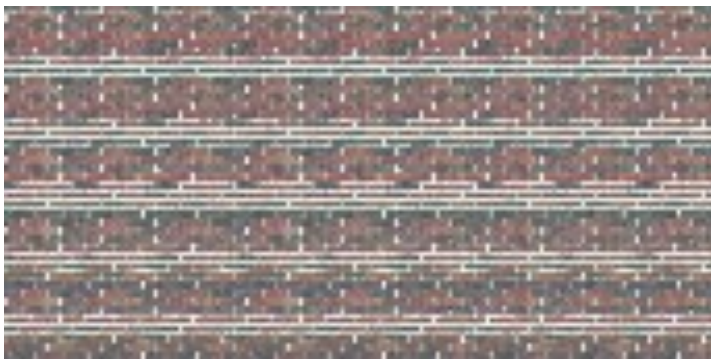
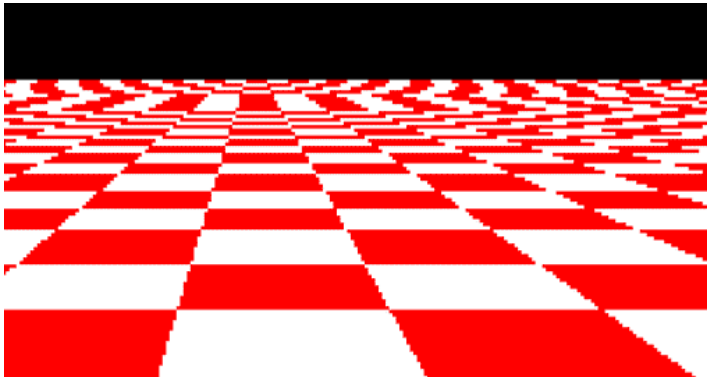
32x32

16x16



Aliasing “in the wild”

Disintegrating textures

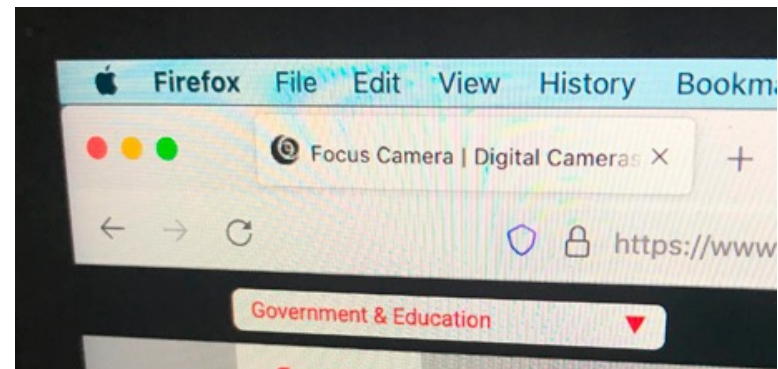


[Source](#)

Moire patterns, false color



[Source](#)



[Source](#)

Aliasing in neural networks

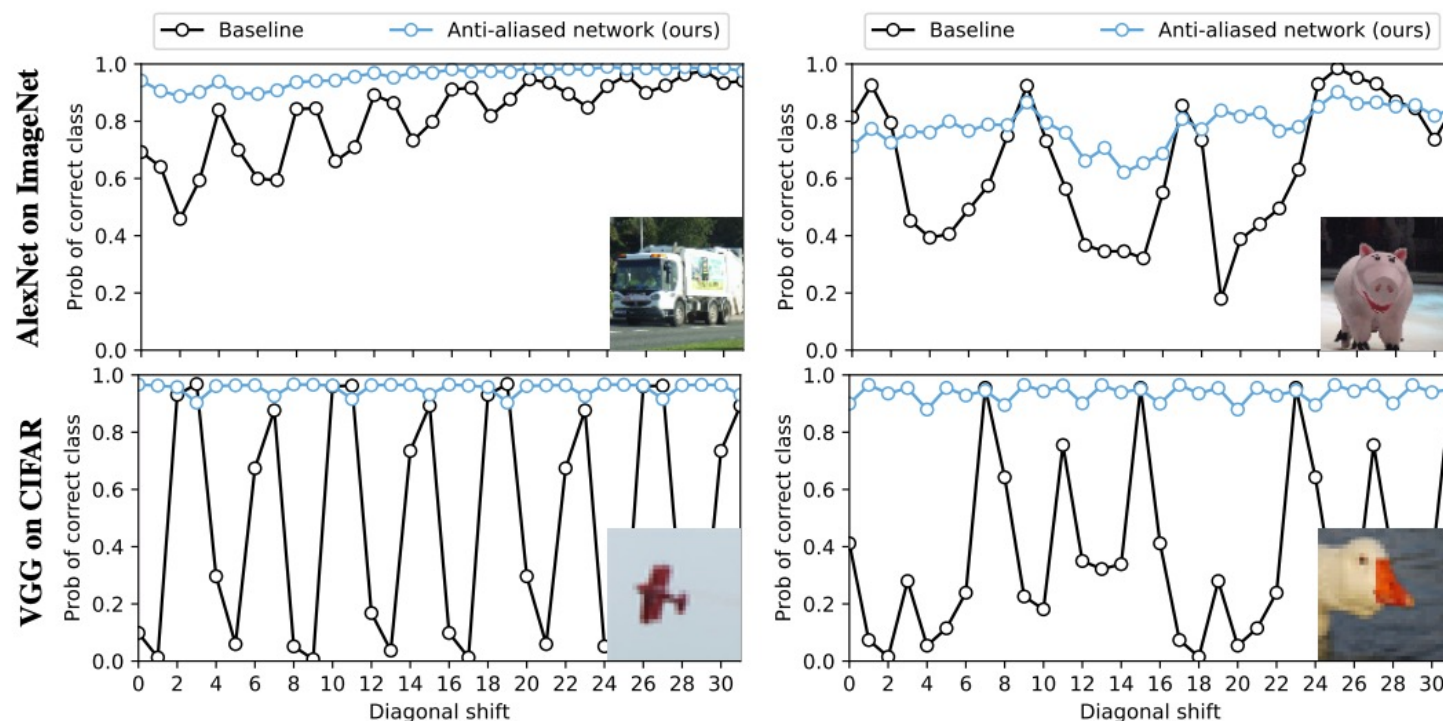
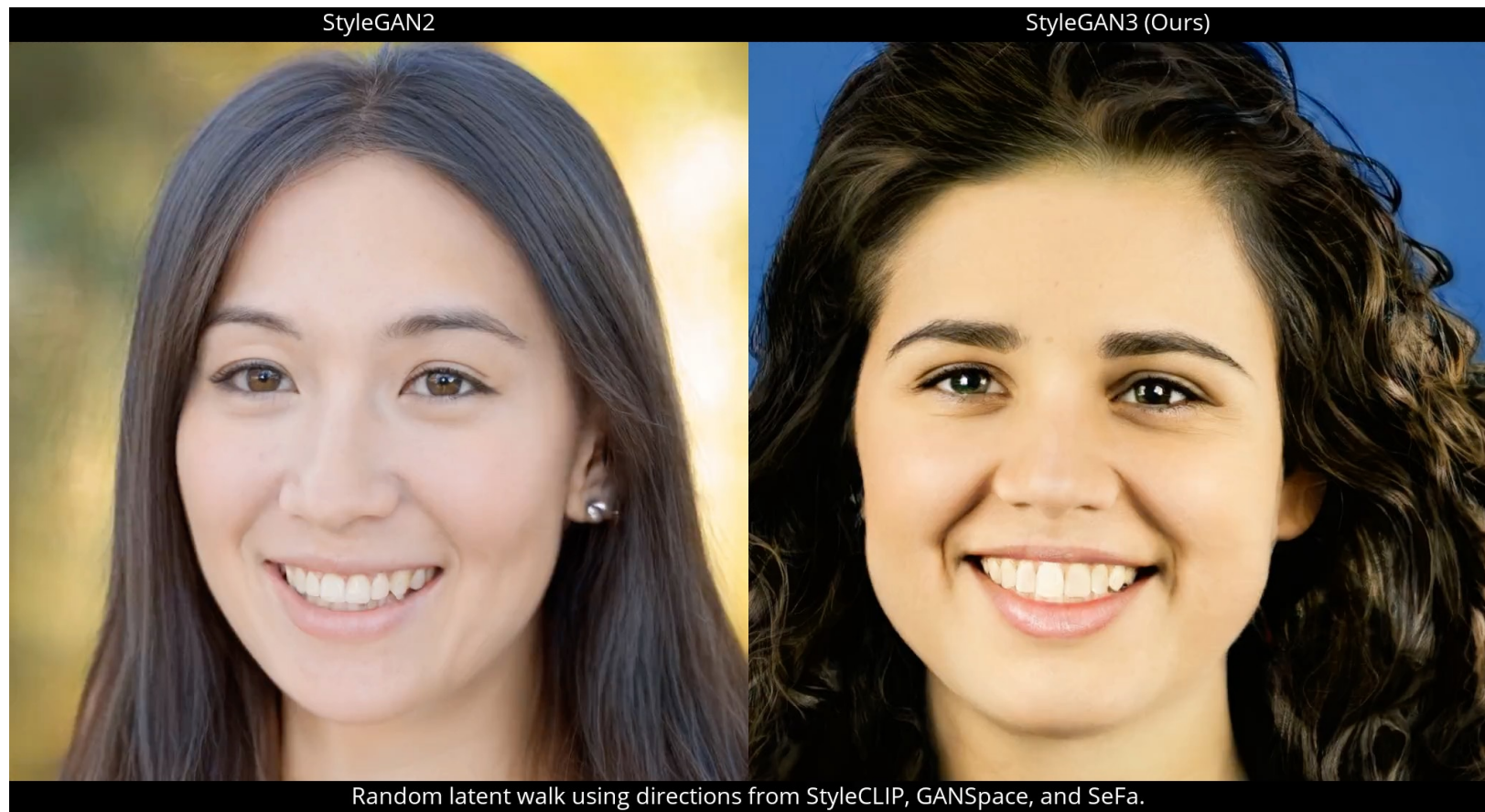


Figure 1. Classification stability for selected images. Predicted probability of the correct class changes when shifting the image. The baseline (black) exhibits chaotic behavior, which is stabilized by our method (blue). We find this behavior across networks and datasets. Here, we show selected examples using AlexNet on ImageNet (**top**) and VGG on CIFAR10 (**bottom**). Code and anti-aliased versions of popular networks are available at <https://richzhang.github.io/antialiased-cnns/>.

R. Zhang. [Making convolutional networks shift-invariant again](#). ICML 2019

Aliasing in neural networks

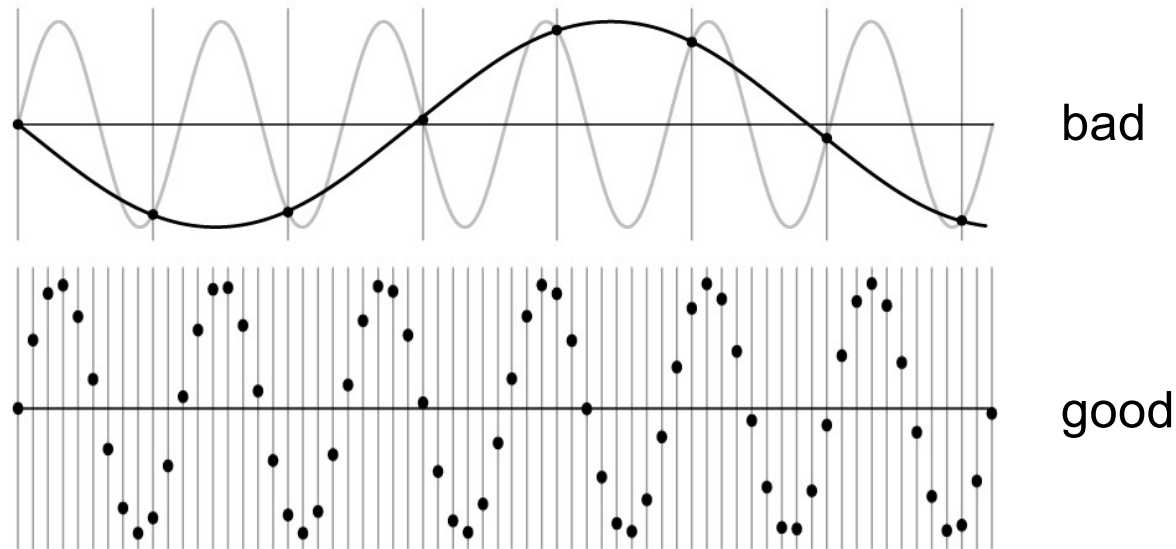


T. Karras et al. [Alias-Free Generative Adversarial Networks](#). NeurIPS 2021

[Videos](#)

Nyquist-Shannon sampling theorem

- When sampling a signal at discrete intervals, the sampling frequency must be at least **twice the maximum frequency of the input signal** to allow us to reconstruct the original perfectly from the sampled version



https://en.wikipedia.org/wiki/Nyquist-Shannon_sampling_theorem

Anti-aliasing

- How can we get rid of aliasing?
 - Sample more often (if you can)
 - Get rid of all frequencies that are greater than half the new sampling frequency
 - Will lose information, but that's better than aliasing
 - How to get rid of high frequencies?
 - Apply a **low-pass filter** (to be covered later)

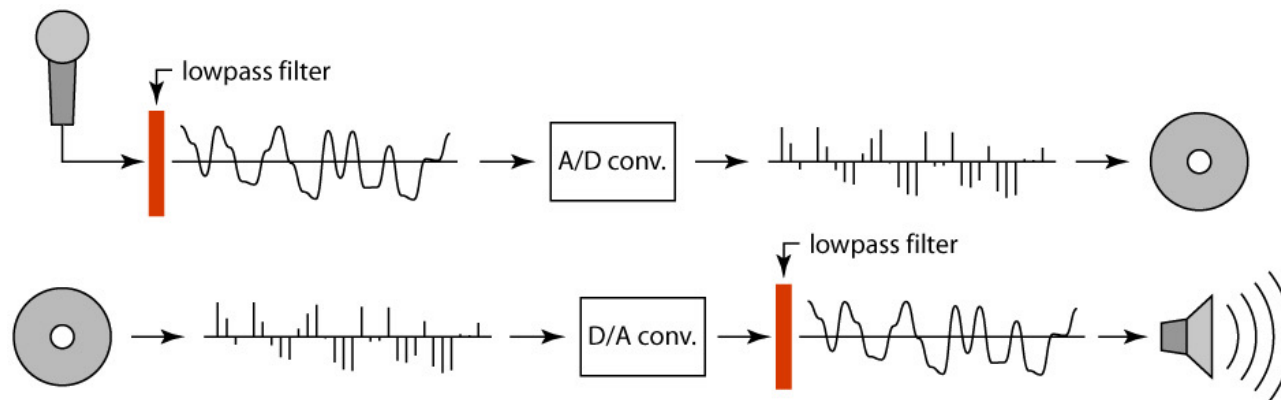
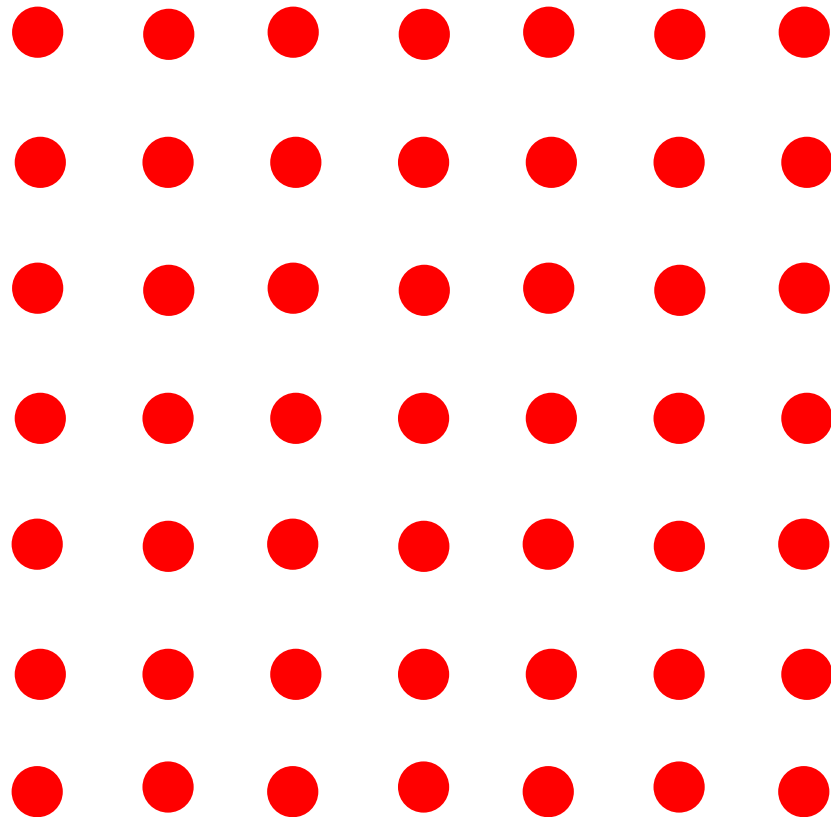


Image processing basics: Outline

- Images as sampled functions
- Sampling and reconstruction, aliasing
- Image resampling, interpolation

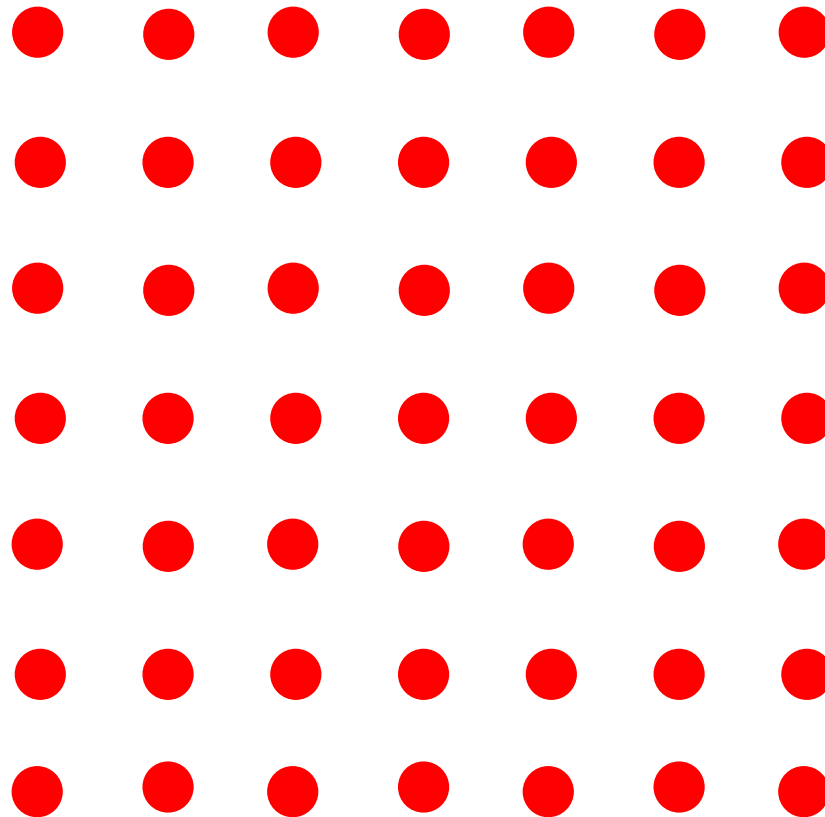
Subsampling an image

- How do we reduce the size of an image by a factor of two?



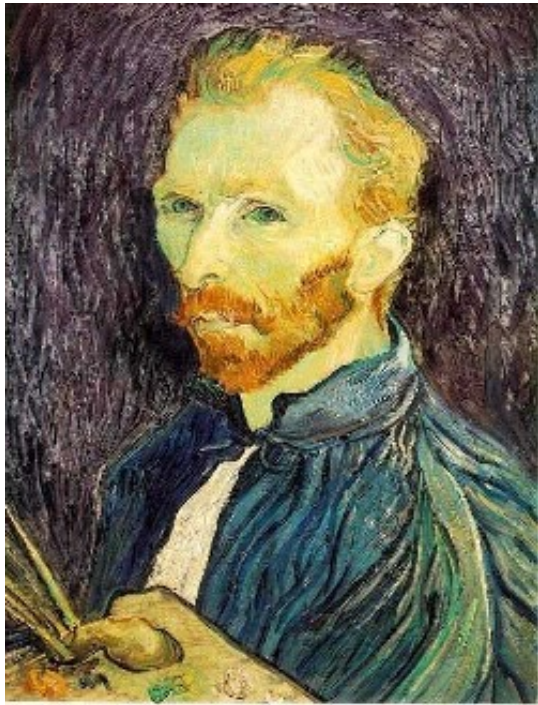
Subsampling an image

- How do we reduce the size of an image by a factor of two?



How about throwing away every other row and column to create a half-size image?

Subsampling without pre-filtering



1/2



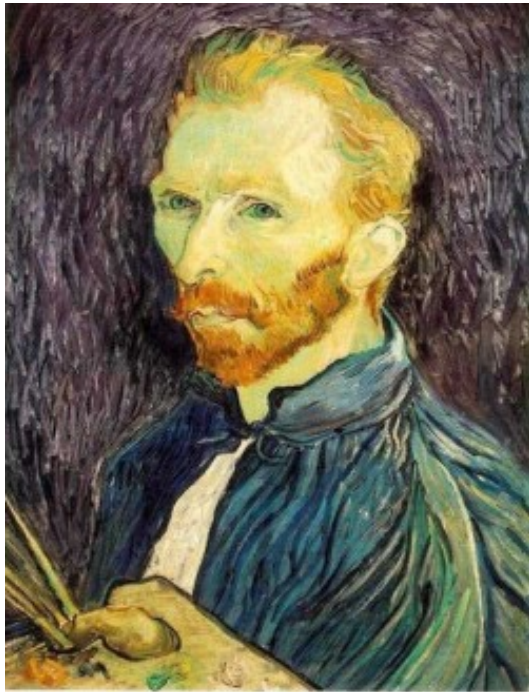
1/4 (2x zoom)



1/8 (4x zoom)

Source: S. Seitz (via D. Hoiem)

Subsampling with pre-filtering



$1/2$



$1/4$



$1/8$

- Image is smoothed with a *Gaussian filter* before subsampling

Source: S. Seitz (via D. Hoiem)

Subsampling with pre-filtering

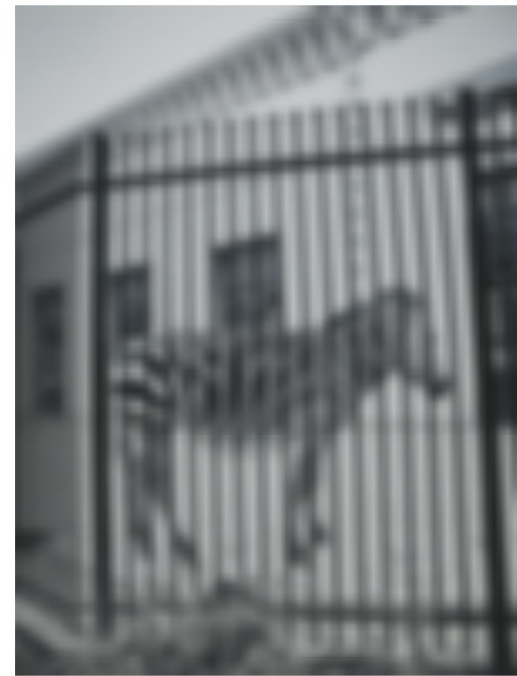
Image



Downsampled by 4



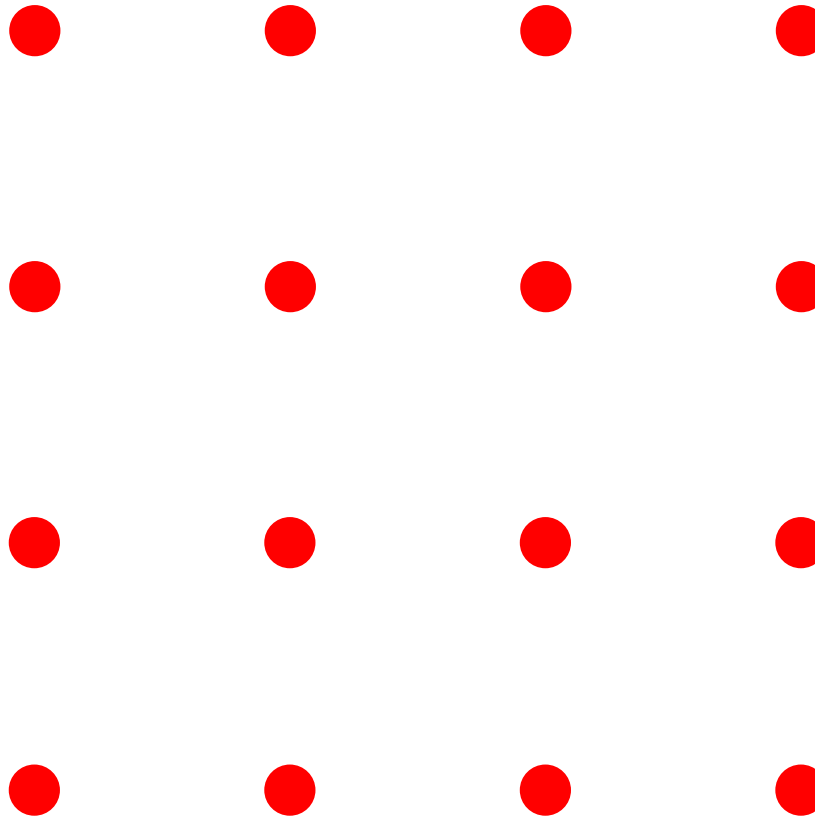
Smoothed then
downsampled by 4



Source: [D. Forsyth](#)

Upsampling an image

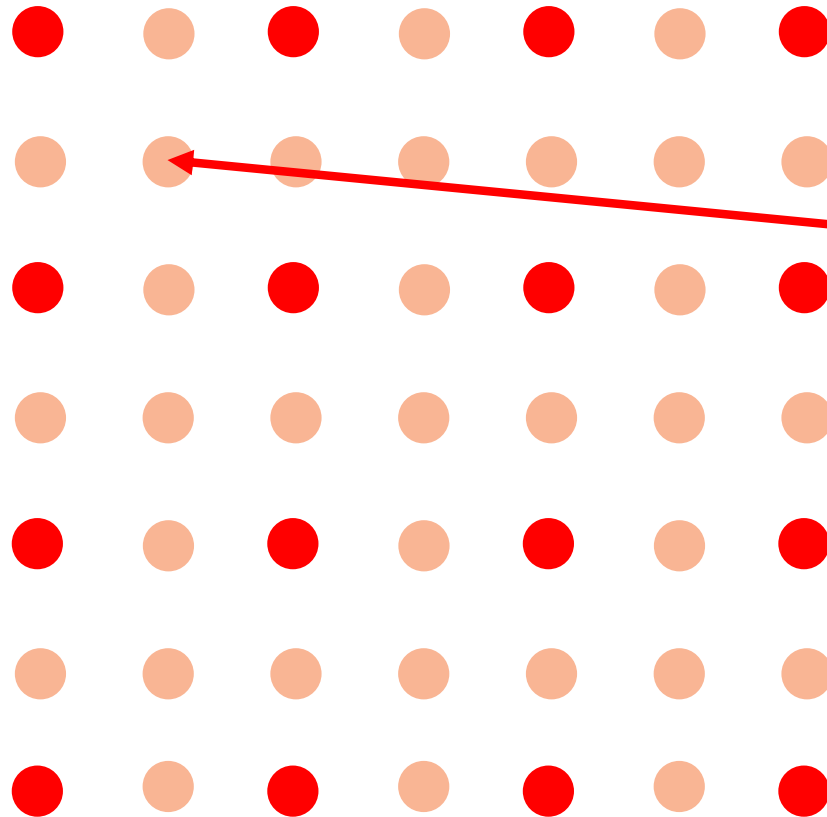
- How do we *increase* the size of an image by a factor of two?



Let's increase the resolution of the sampling grid!

Upsampling an image

- How do we *increase* the size of an image by a factor of two?

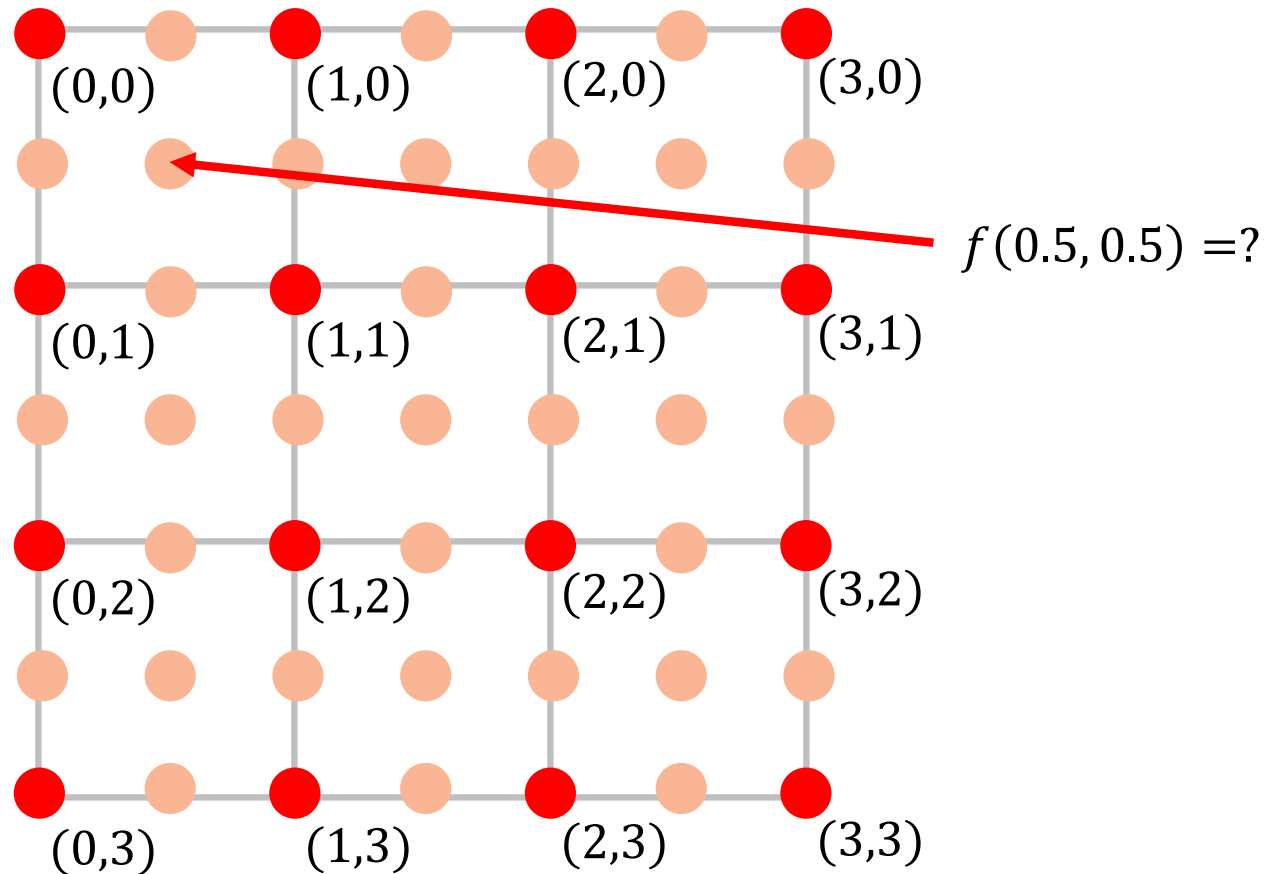


What should this
value be?

Need to *interpolate*!

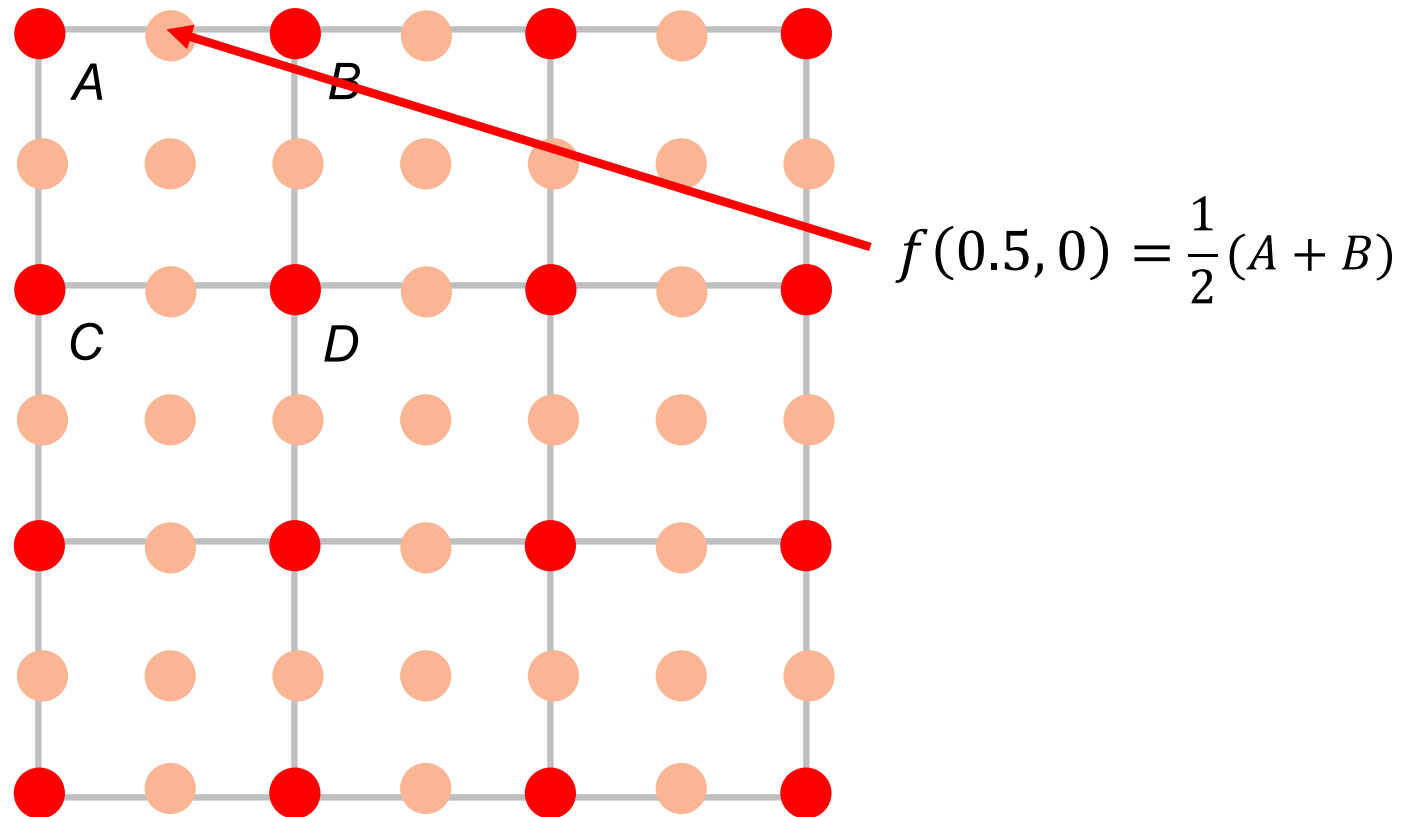
Upsampling an image

- How do we *increase* the size of an image by a factor of two?



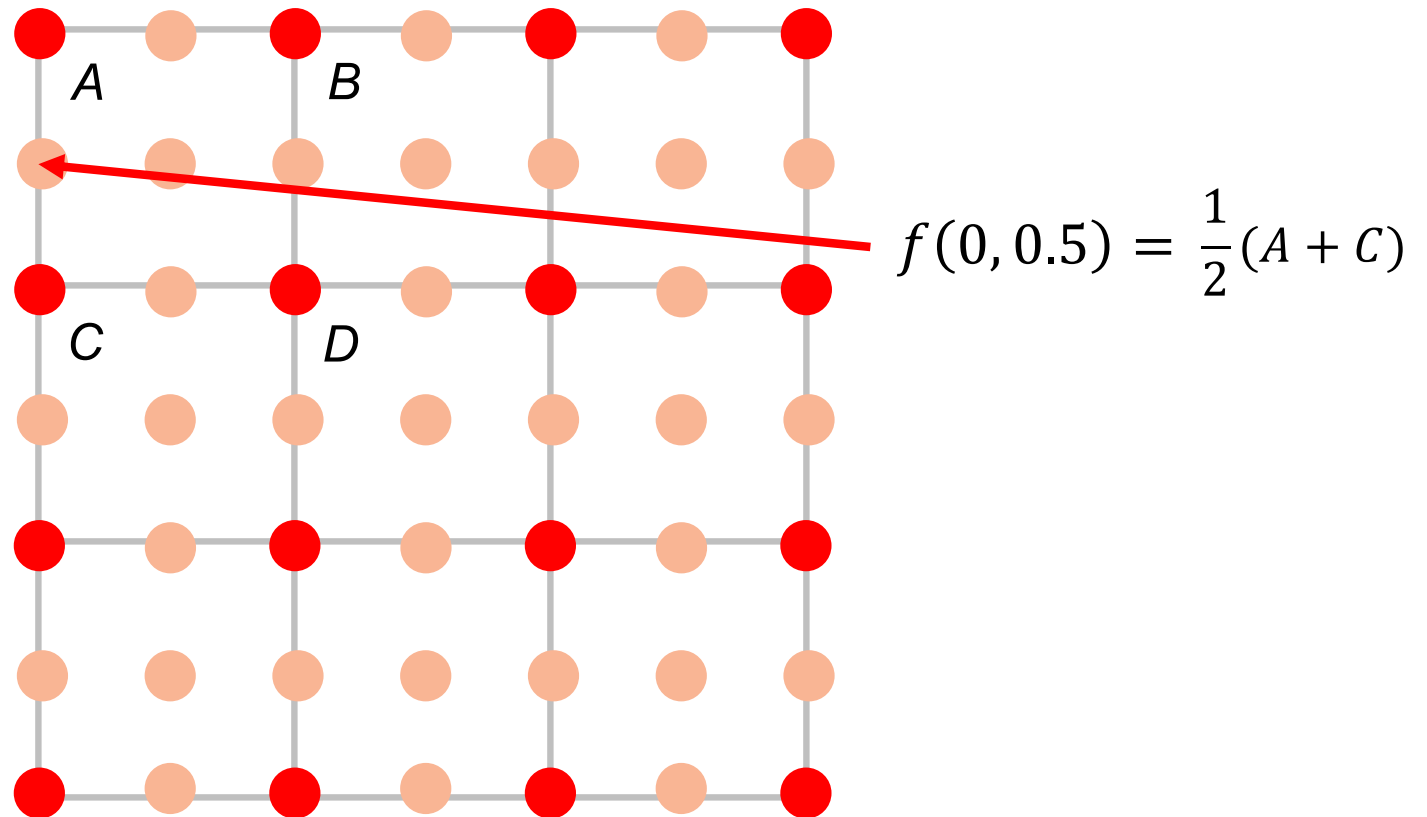
Bilinear interpolation

- Let $f(0, 0) = A, f(1, 0) = B, f(0, 1) = C, f(1, 1) = D$



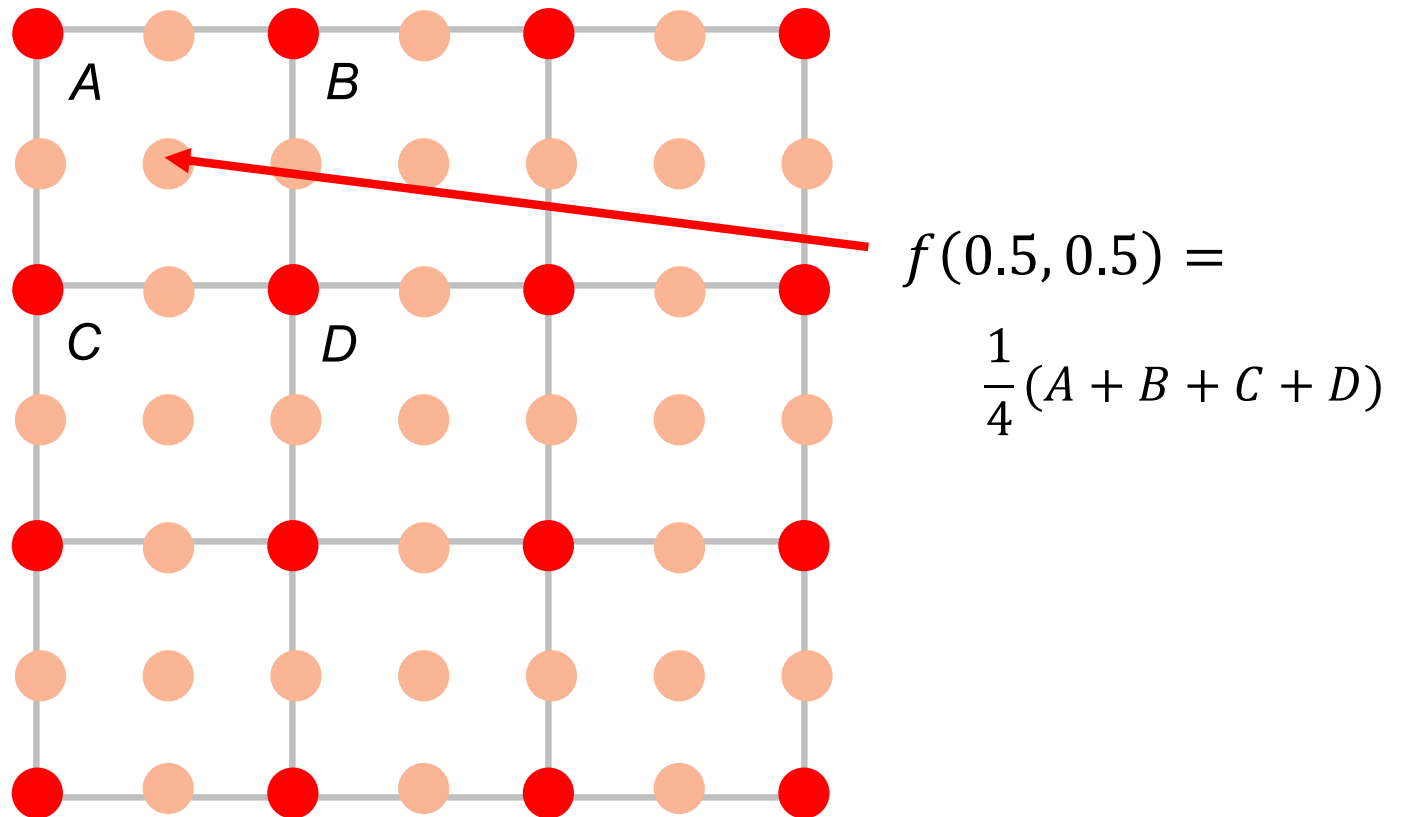
Bilinear interpolation

- Let $f(0, 0) = A, f(1, 0) = B, f(0, 1) = C, f(1, 1) = D$

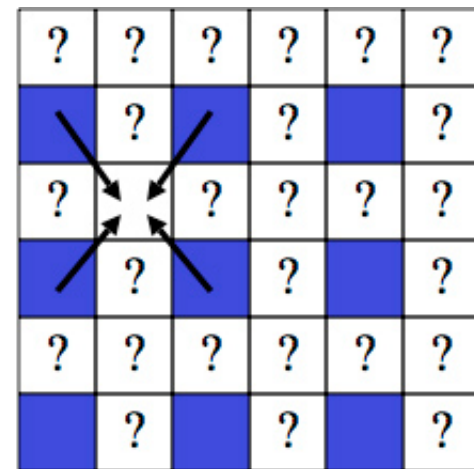
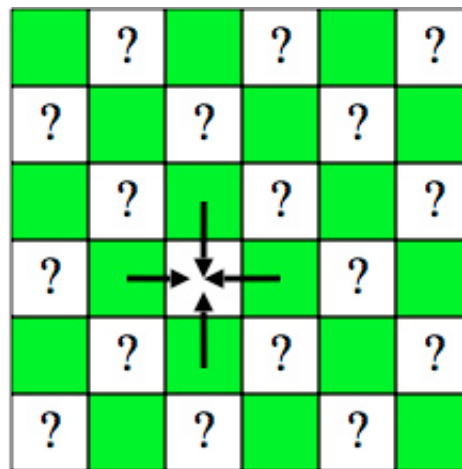


Bilinear interpolation

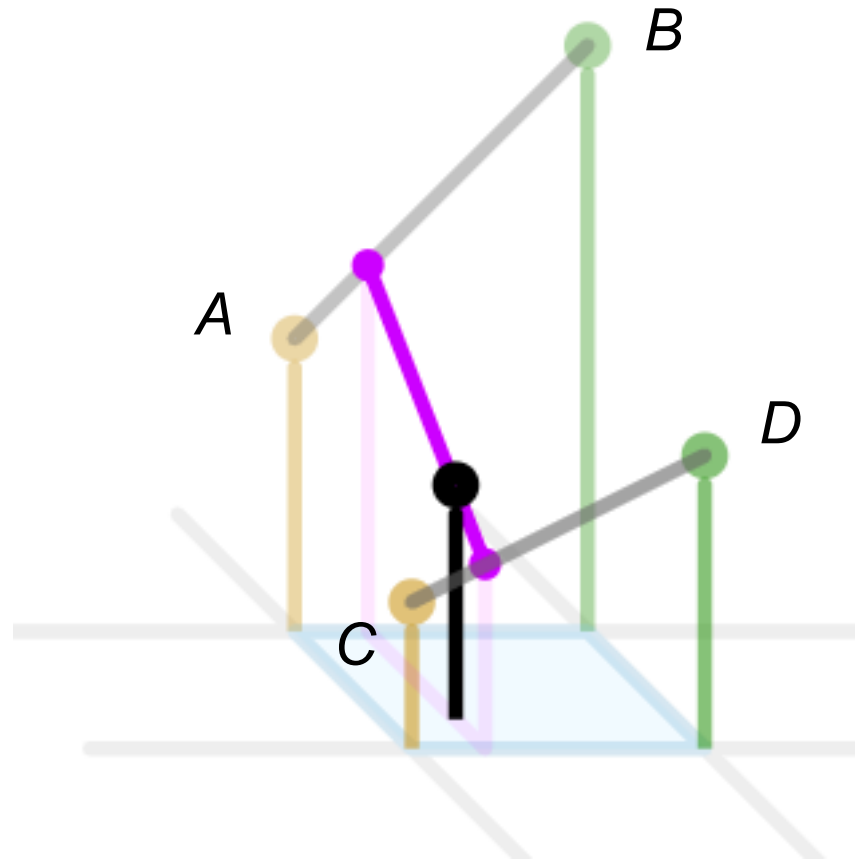
- Let $f(0, 0) = A, f(1, 0) = B, f(0, 1) = C, f(1, 1) = D$



Application: Demosaicing

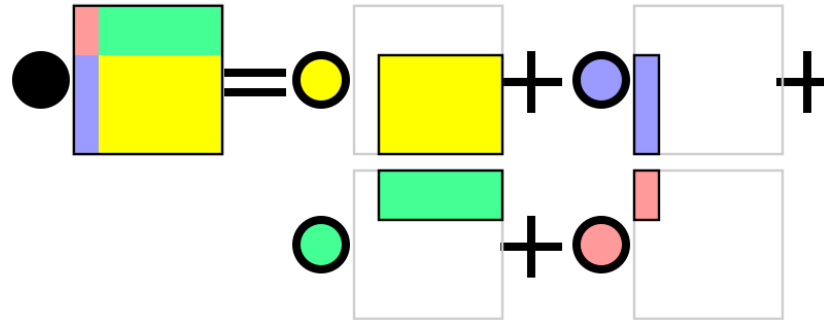
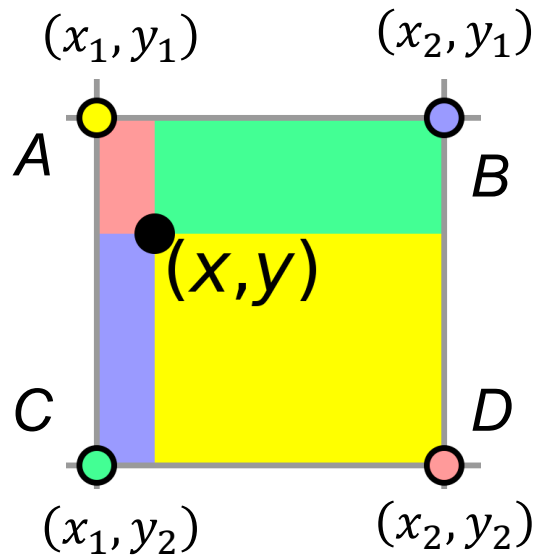


Bilinear interpolation more generally



http://en.wikipedia.org/wiki/Bilinear_interpolation

Bilinear interpolation more generally



$$f(x, y) = w_{11}A + w_{21}B + w_{12}C + w_{22}D$$

$$w_{11} = \frac{(x_2 - x)(y_2 - y)}{(x_2 - x_1)(y_2 - y_1)}$$

$$w_{12} = \frac{(x_2 - x)(y - y_1)}{(x_2 - x_1)(y_2 - y_1)}$$

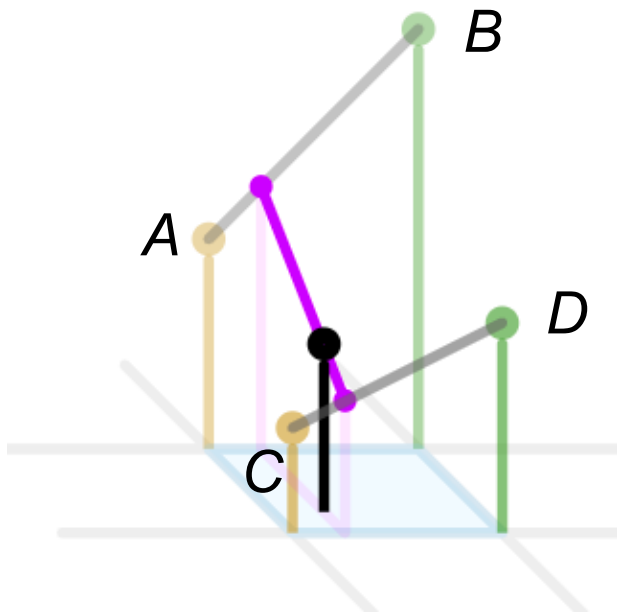
$$w_{21} = \frac{(x - x_1)(y_2 - y)}{(x_2 - x_1)(y_2 - y_1)}$$

$$w_{22} = \frac{(x - x_1)(y - y_1)}{(x_2 - x_1)(y_2 - y_1)}$$

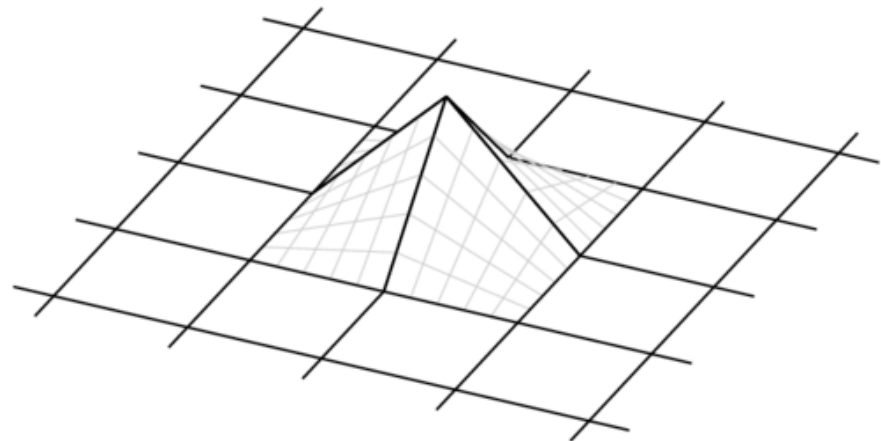
http://en.wikipedia.org/wiki/Bilinear_interpolation

Bilinear interpolation: Basis function view

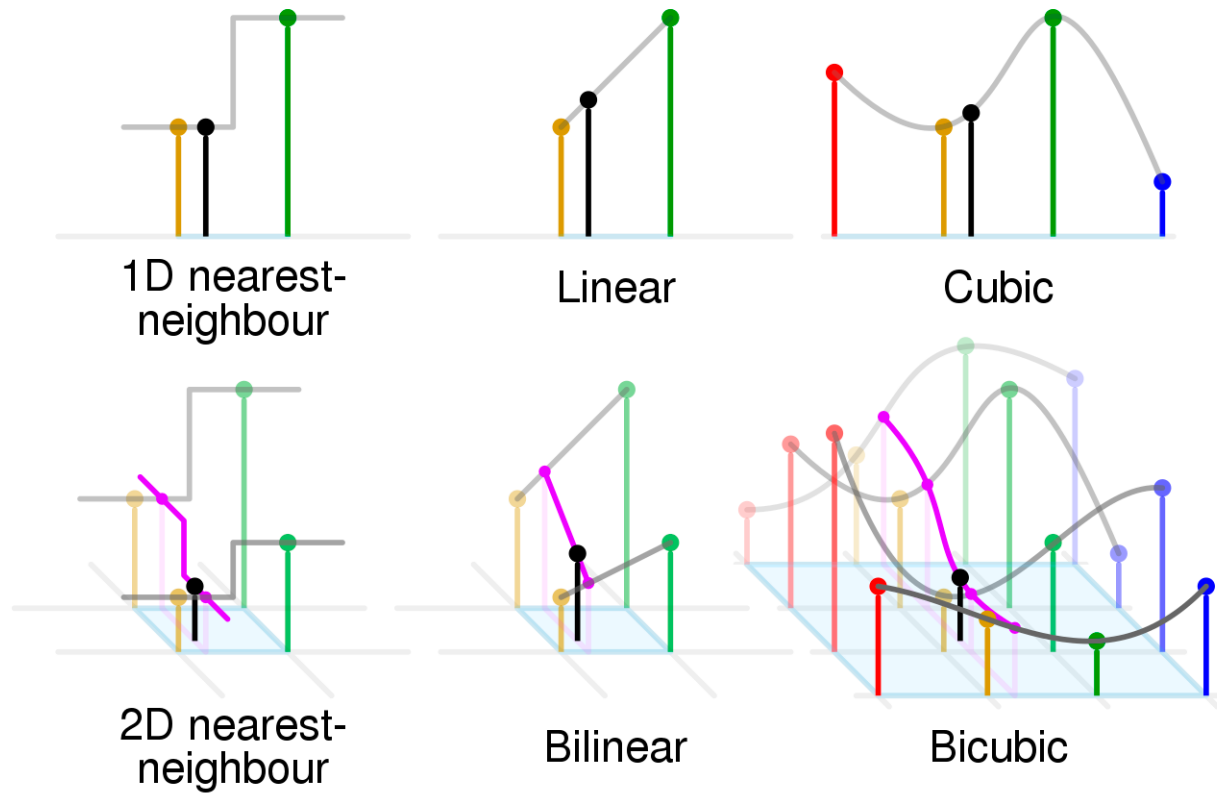
- Interpolated function is sum of basis functions or “bumps” centered at the four adjacent grid points, weighted by the image values at the corresponding points



Bilinear basis function



Other kinds of interpolation



Source: [Wikipedia](https://en.wikipedia.org/wiki/Interpolation)

Interpolation and function extrema

- When you use linear interpolation, extrema of the image function can only occur at the original sample points
- What about nonlinear interpolation?

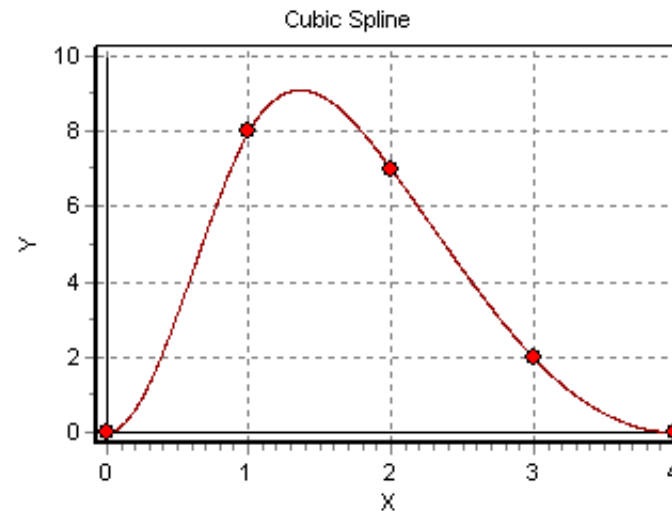
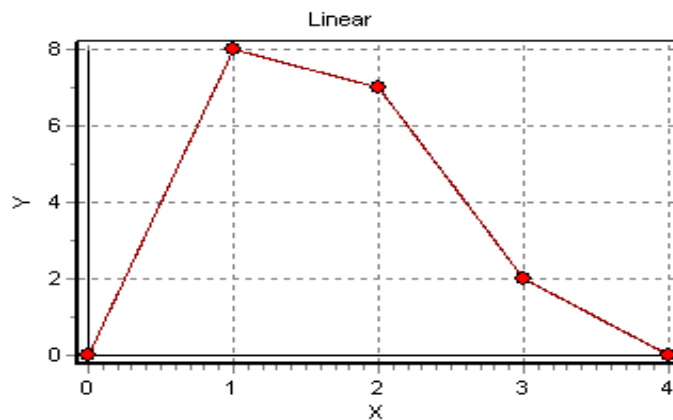
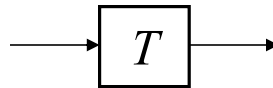


Image processing basics: Outline

- Images as sampled functions
- Sampling and reconstruction, aliasing
- Image resampling, interpolation
- Image transformations

Image transformations



Downsampling



Image transformations

Upsampling

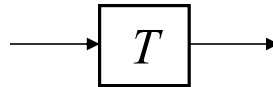


Image transformations

Contrast change

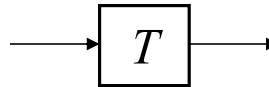


Image transformations

Blurring/sharpening

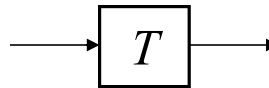


Image transformations

Warping

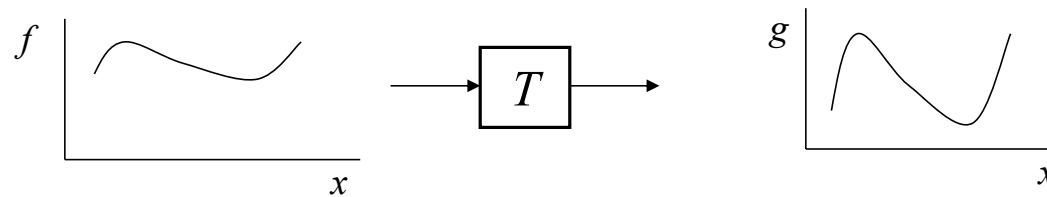


Last time: Image processing basics

- Images as sampled functions
- Sampling and reconstruction, aliasing
- Image resampling, interpolation
- Image transformations

Point processing

- Change **range** of image: $g = T(f)$



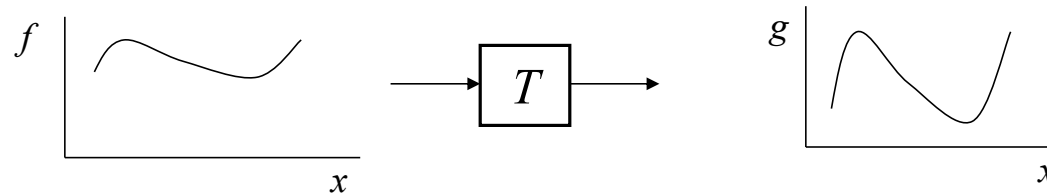
- Negative: $g = 1 - f$



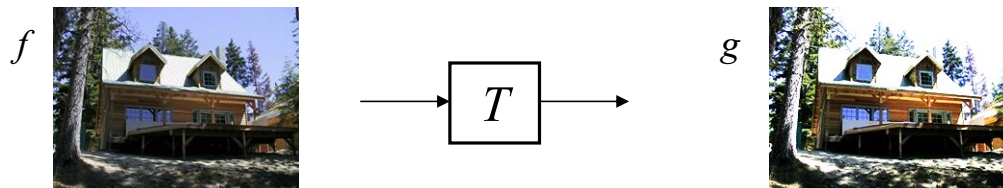
[Image source](#)

Point processing

- Change **range** of image: $g = T(f)$

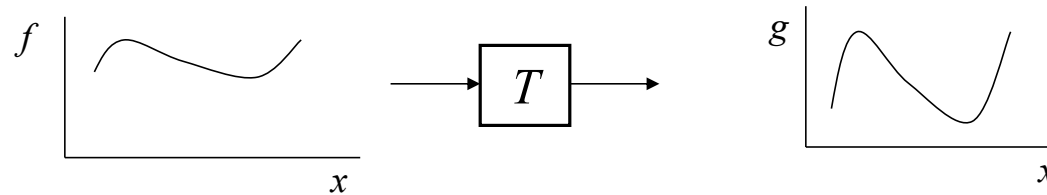


- Affine* contrast adjustment: $g = af + b$

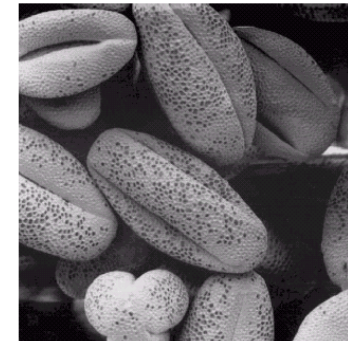
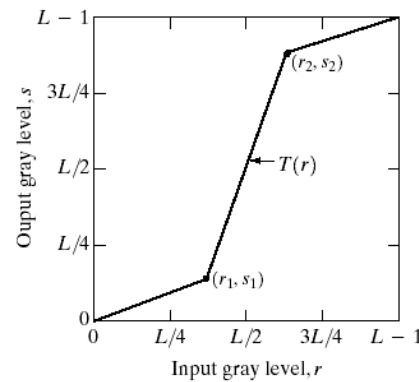
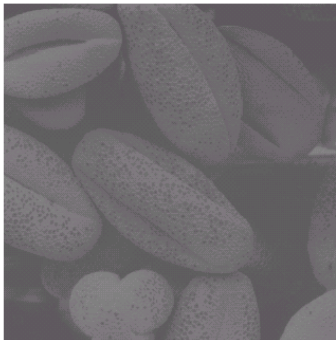


Point processing

- Change **range** of image: $g = T(f)$

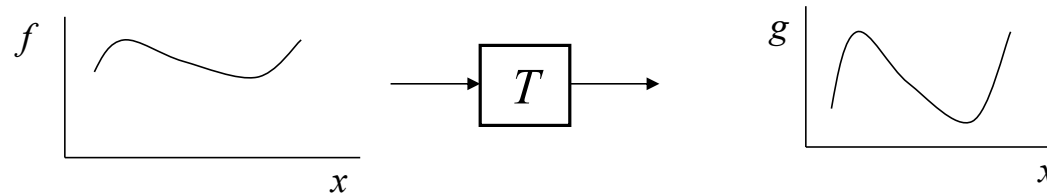


- Piecewise-linear* contrast adjustment:



Point processing

- Change **range** of image: $g = T(f)$



- Gamma correction: $g = af^\gamma$



[Source](#)

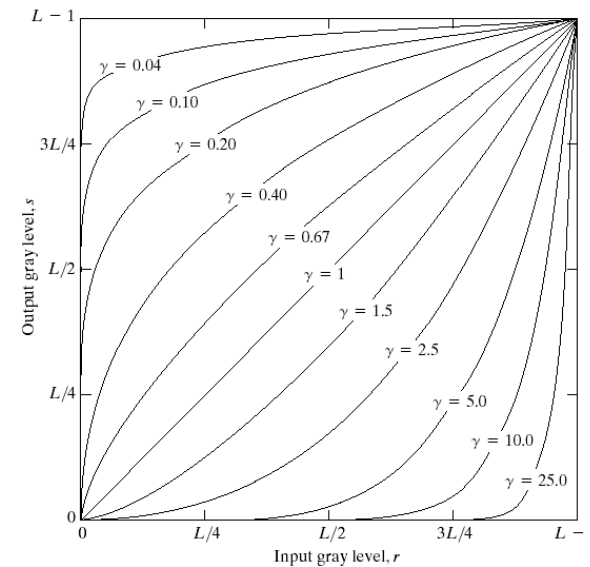


Image filtering

- Roughly speaking, replace image value at x with some function of values in its spatial neighborhood $N(x)$:

$$g(x) = T(f(N(x)))$$

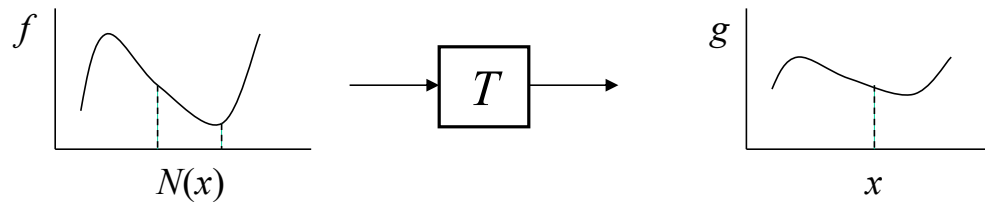
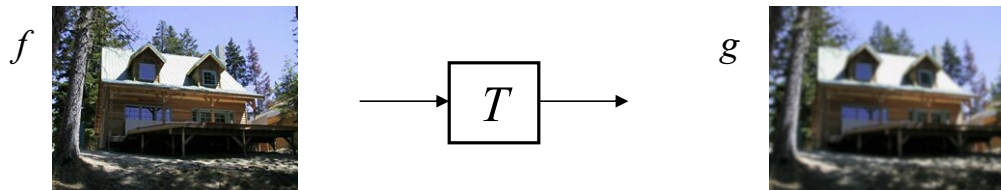


Image filtering

- Roughly speaking, replace image value at x with some function of values in its spatial neighborhood $N(x)$:

$$g(x) = T(f(N(x)))$$



- Examples: smoothing, sharpening, edge detection, etc.

Image warping

- Change **domain** of image:

$$x' = T(x), \quad g(x') = f(x)$$

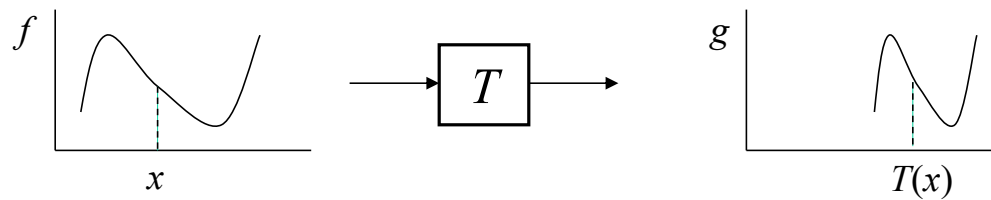


Image warping

- Change **domain** of image:

$$x' = T(x), \quad g(x') = f(x)$$

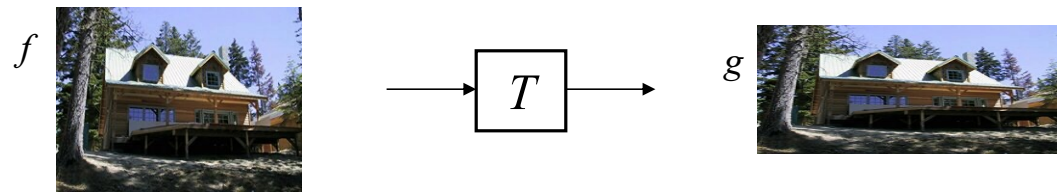


Image warping

- Examples of *global parametric* warps:



translation



rotation



scaling (uniform or
non-uniform)



affine



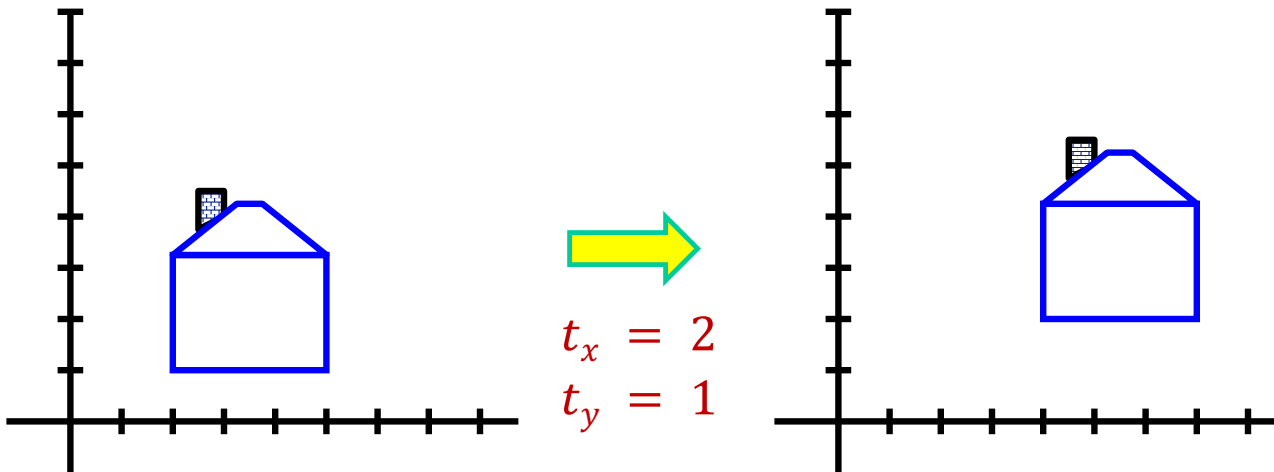
perspective
homography



cylindrical
(not covered
in this class)

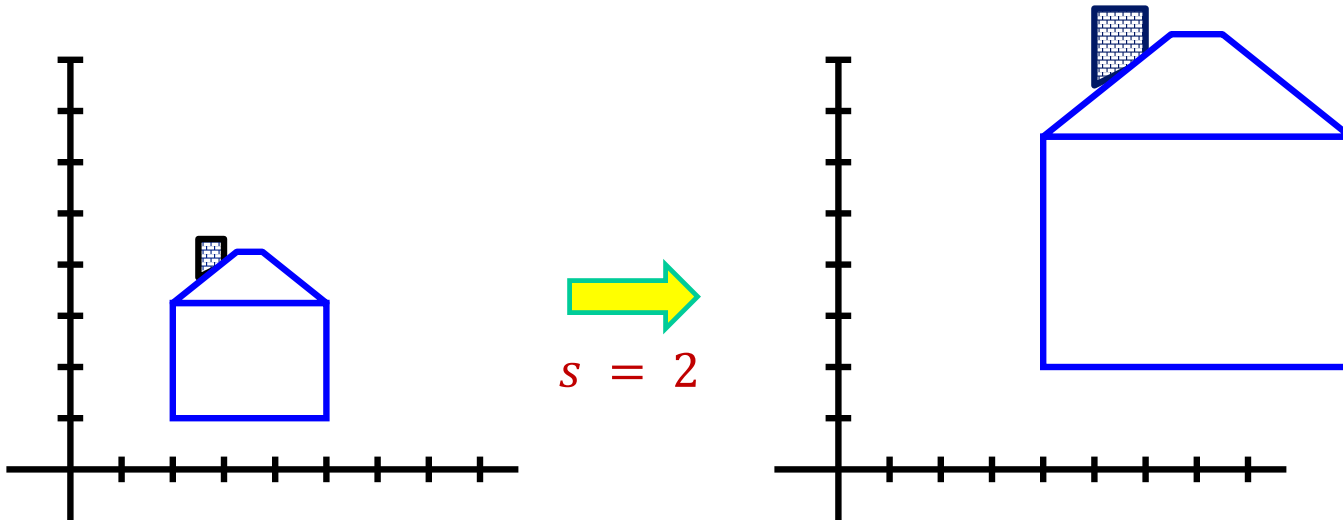
Translation

$$\begin{aligned}x' &= x + t_x \\ y' &= y + t_y\end{aligned}$$



Uniform scaling

$$\begin{aligned}x' &= s * x \\ y' &= s * y\end{aligned}$$

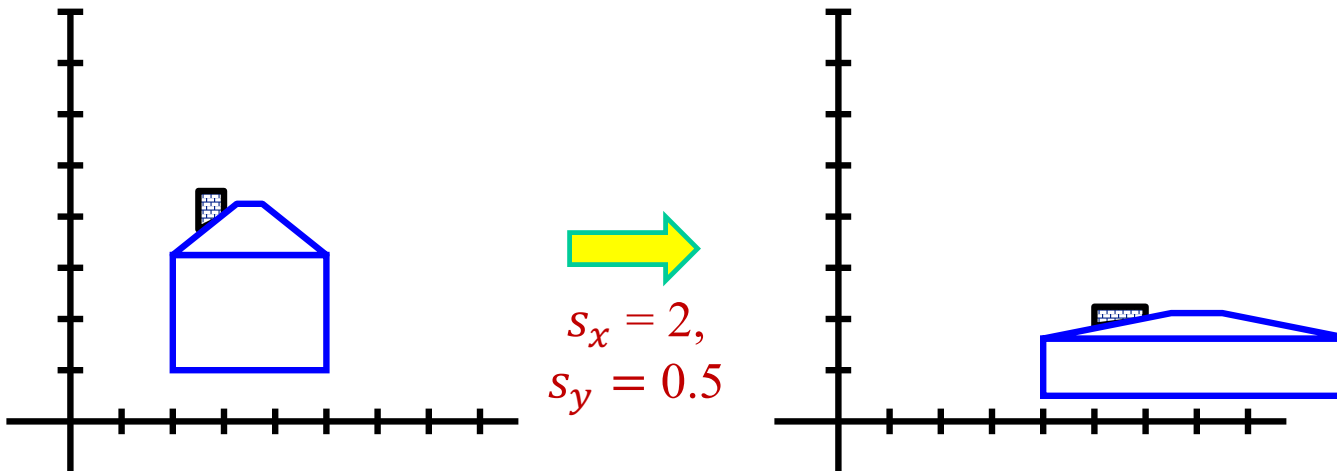


Non-uniform scaling

In matrix form:

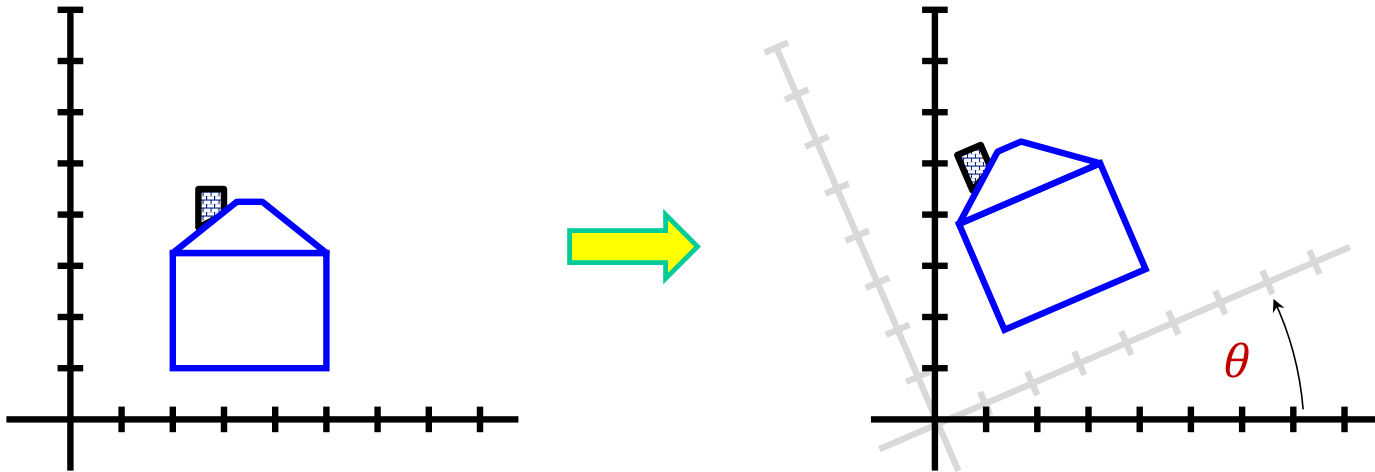
$$\begin{aligned}x' &= s_x * x \\ y' &= s_y * y\end{aligned}$$

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$



Rotation

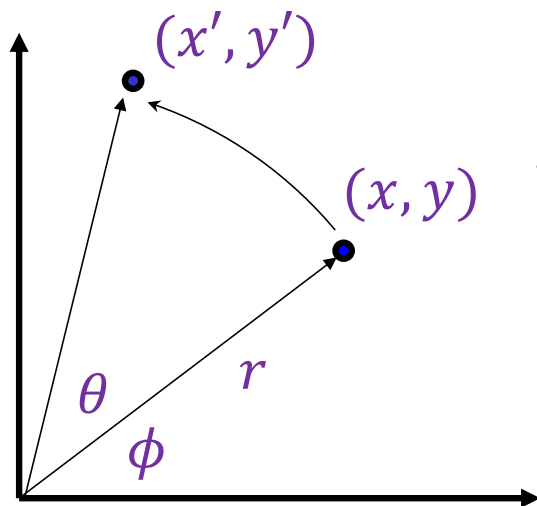
- Rotate the image by an angle of θ about the origin:



Rotation

- Rotate the image by an angle of θ about the origin:

$$\begin{aligned}x' &= r \cos(\phi + \theta) = r \cos(\phi) \cos(\theta) - r \sin(\phi) \sin(\theta) = x \cos(\theta) - y \sin(\theta) \\y' &= r \sin(\phi + \theta) = r \sin(\phi) \cos(\theta) + r \cos(\phi) \sin(\theta) = x \sin(\theta) + y \cos(\theta)\end{aligned}$$



Apply trig identities

Substitute

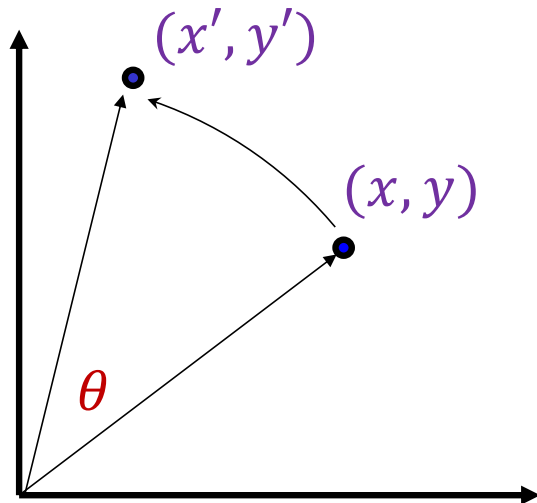
$$\begin{aligned}x &= r \cos(\phi) \\y &= r \sin(\phi)\end{aligned}$$

Rotation

- Rotate the image by an angle of θ about the origin:

$$x' = x \cos(\theta) - y \sin(\theta)$$

$$y' = x \sin(\theta) + y \cos(\theta)$$



In matrix form:

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

Rotation

- 2D rotation in matrix form:

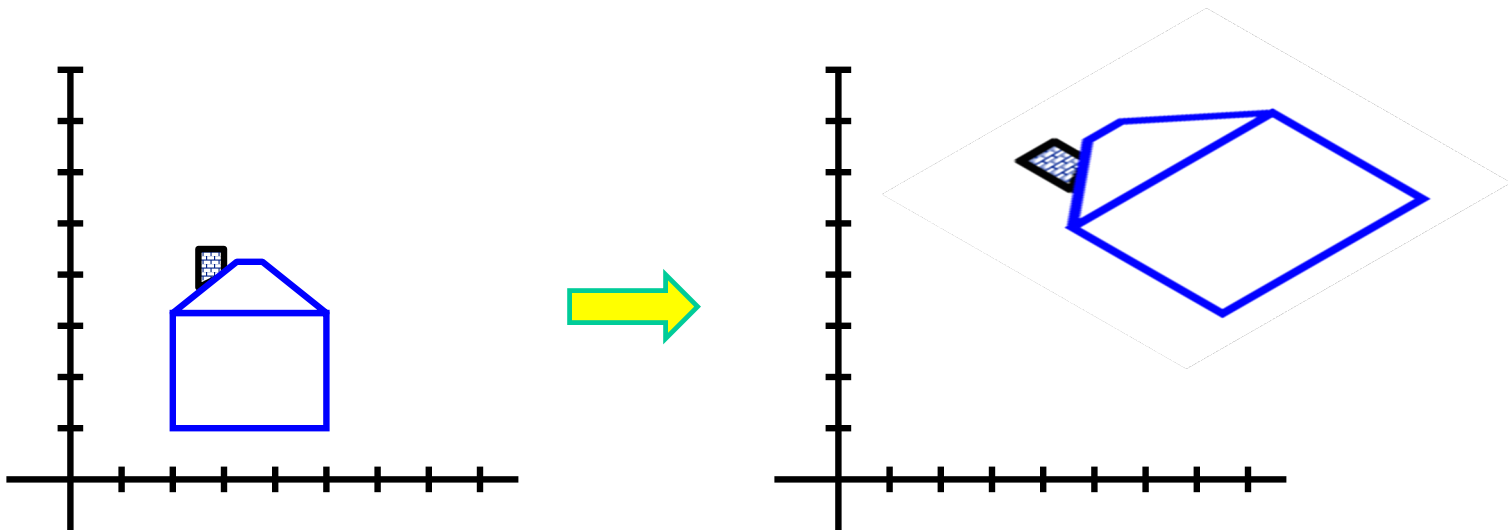
$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \underbrace{\begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}}_{R(\theta)} \begin{pmatrix} x \\ y \end{pmatrix}$$

- Note: even though $\cos(\theta)$ and $\sin(\theta)$ are *nonlinear* functions of θ , x' and y' are *linear* combinations of x and y
- What is the inverse transformation?
 - Rotation by $-\theta$
 - For rotation matrices, $R^{-1} = R^T$

Affine transformation

- Combination of translation, non-uniform scaling, rotation, and *shear*

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} e \\ f \end{pmatrix}$$

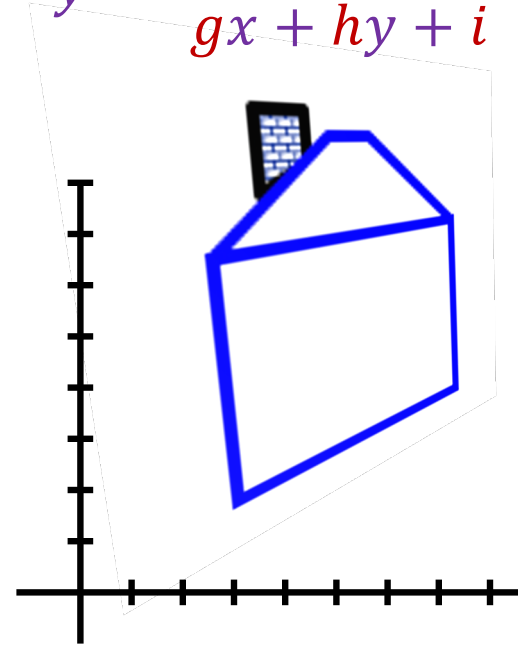
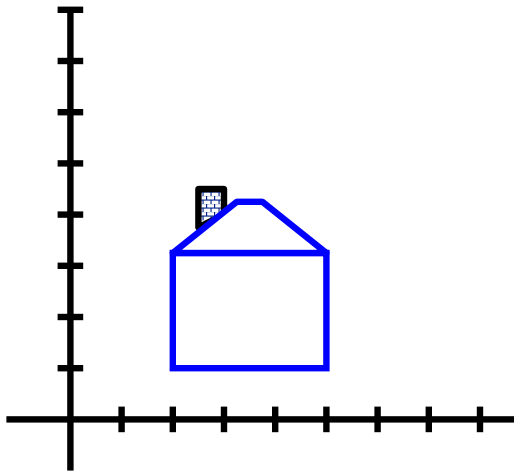


Projective homography

- A transformation that preserves straight lines, but not parallelism

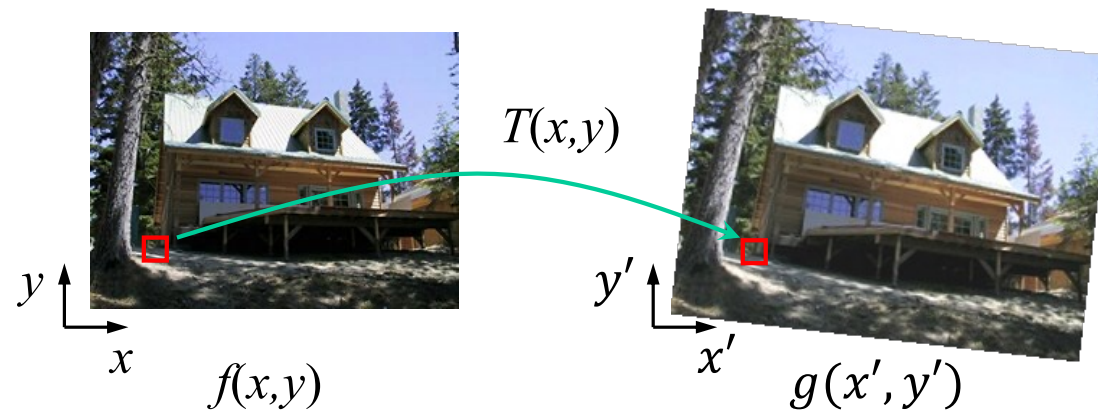
$$x' = \frac{ax + by + c}{gx + hy + i},$$

$$y' = \frac{dx + ey + f}{gx + hy + i}$$



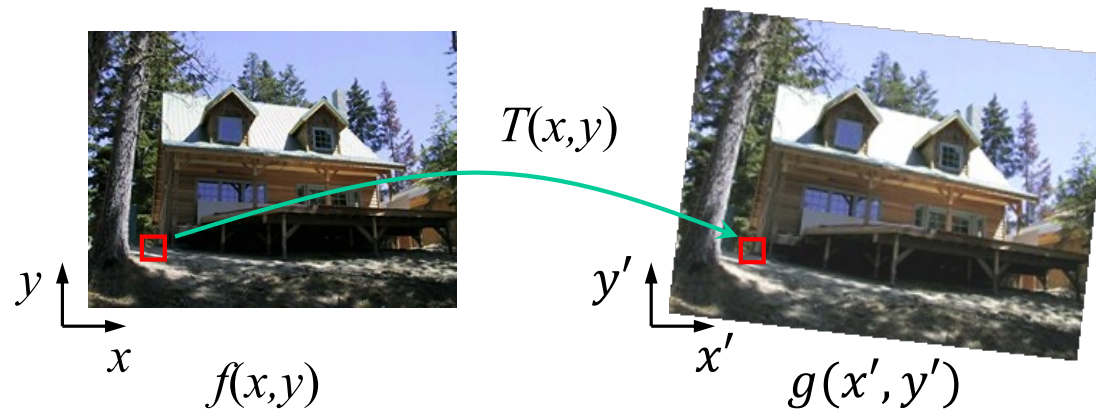
Warping pixels

- Given a coordinate transform $(x', y') = T(x, y)$ and a source image $f(x, y)$, how do we compute a transformed image $g(x', y') = f(T(x, y))$?



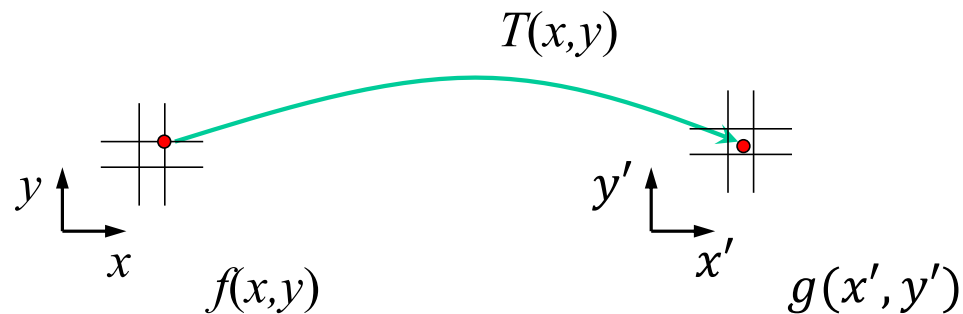
Forward warping

- Send each pixel $f(x, y)$ to its corresponding location $(x', y') = T(x, y)$ in the second image



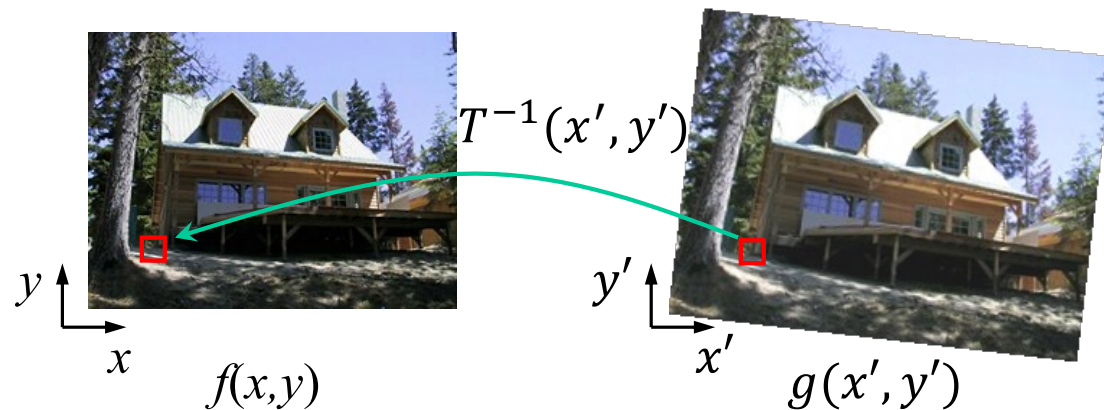
Forward warping

- Send each pixel $f(x, y)$ to its corresponding location $(x', y') = T(x, y)$ in the second image
 - What if (x', y') is not an integer location on the pixel grid?
 - We can “distribute” colors among the neighboring grid locations – known as *splatting*



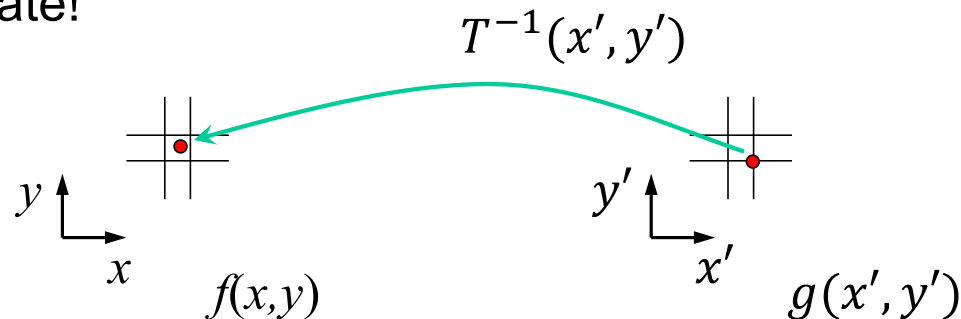
Inverse warping

- For each pixel grid location (x', y') in the second image, get the color from its corresponding location $(x, y) = T^{-1}(x', y')$ in the first image



Inverse warping

- For each pixel grid location (x', y') in the second image, get the color from its corresponding location $(x, y) = T^{-1}(x', y')$ in the first image
 - What if (x, y) is not an integer location on the original pixel grid?
 - Interpolate!



- Inverse warping is usually preferred to forward warping: more efficient, no holes
- However, it requires an invertible warp function, which is not always possible